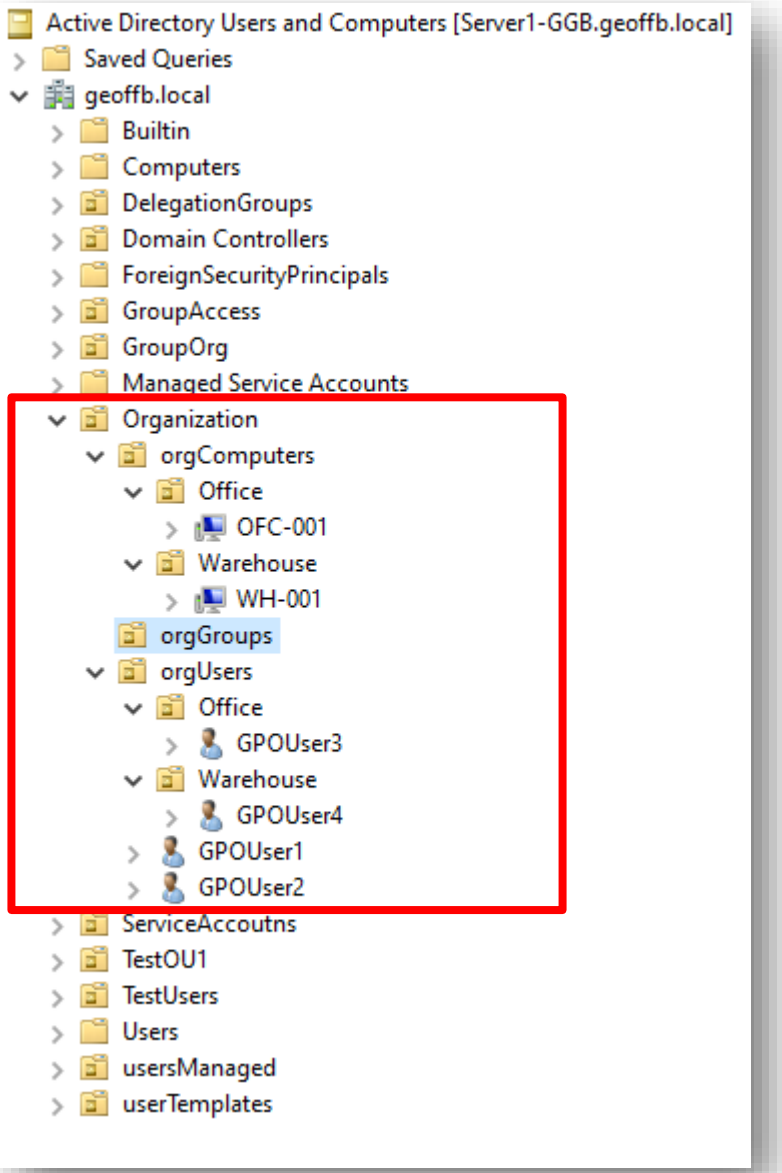
Do this…

1. Create an OU called Organization
   a) Create a subOU called Organization/orgUsers.
2. In Organization/orgUsers
   a) Create two SubOUs Called orgUsers/Office and orgUsers/Warehouse
   b) Create 4 users: GPOUser1, GPOUser2, GPOUser3, GPOUser4.
   c) Move users GPOUser1, and GPOUser2 to Organization/orgUsers
   d) Move GPOUser3 to orgUsers/Office; Move GPOUser4 to orgUsers/Warehouse
3. Create an OU called Organization/orgComputers
   a) Create TWO SubOUs called orgComputers/Office and orgComputers/Warehouse
   b) In orgComputers/Office create a computer called OFC-001
   c) In orgComputers/Warehouse create a computer called WH-001
4. Create an OU called Organization/orgGroups
   a) Create two SubOUs called orgGroups/userGroups and orgGroups/accessGroups



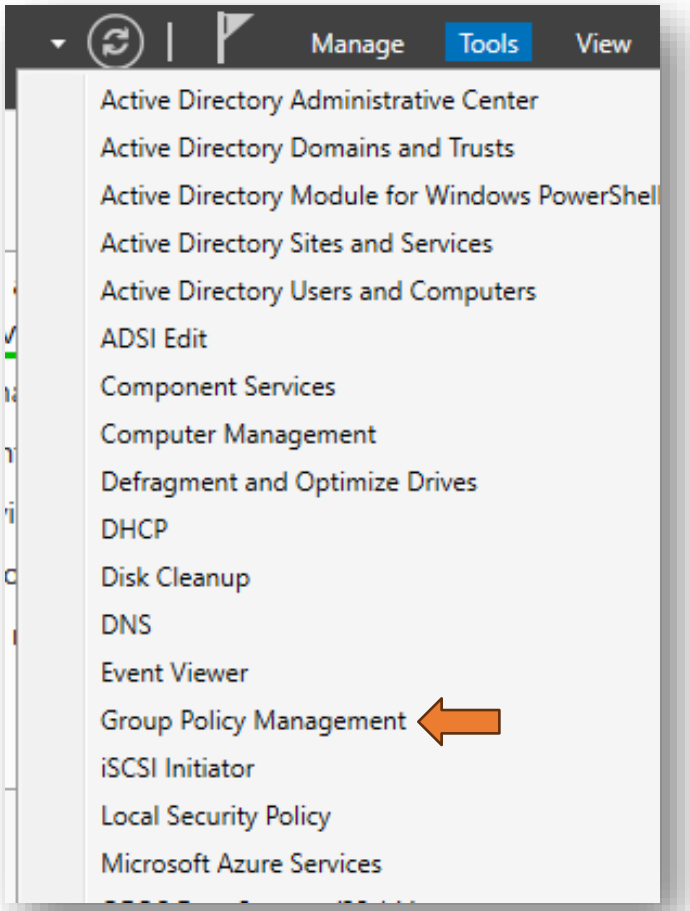
Intro to Group Policy

Module 4

Pages 702-722

What is Group Policy?

- Group policy (GP) is used in a domain to consistently configure, and secure computers; and to standardizing the work environment.
- GP consists of Group Policy Objects (GPO).
  - An **object** that holds a collection of settings that define what a system will look like and how it will behave
- Created, configured and managed through Group Policy Management Console (GPMC.msc).
  - Opened on a domain controller using server manager → tools
  - MMC
  - run → gpmc.msc
- Using GPMC, GPOs can be applied to the Site, the Domain, or the OU containers within an AD Domain.
- Local Group Policy edited by opening GPEdit.msc
  - Only applies to local group policy.



Group Policy (characteristics)

You've done this

- Defining characteristics of Group Policy:
  - Group Policy can be "linked" to a site, domain, OU, or local computer with the OU is the *smallest* AD container.
  - Group Policy *cannot* be linked to non-OU containers
    - Built-in, Computers, ForeignSecurityPricipals, Managed Services Accounts, Users
  - Group Policy settings are found in GPOs and those GPOs are stored in the Group Policy Objects container in GPMC.
    - GPOs have a unique name and GUID
    - 2 GPOs are created in AD by default:
      - Default Domain Policy
      - Default Domain Controller Policy

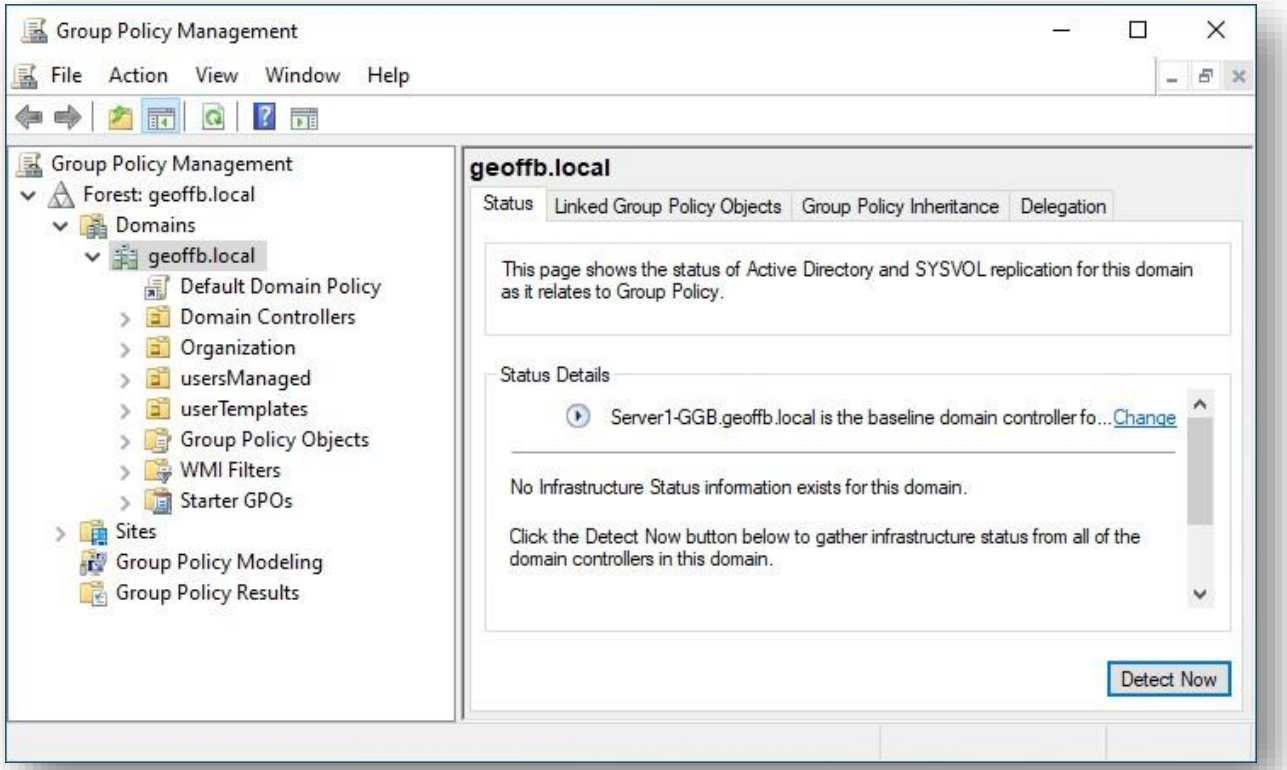
Group Policy (characteristics)

- GPOs can be local and non-local
  - ***Local*** is applied to the computer (rather than applied *at* the computer).
  - ***Non-local*** is applied to the Site, Domain and OU.
- Group Policy can be configured to affect user accounts and computers
  - Affects user configuration, computer configuration or both.
  - The computer configuration will override the user configuration (most of the time) if there is a conflict while both are in effect.
- When Group Policy is updated, old policies are removed or updated for all clients
  - Default GP update time is 90 minutes or when the computer starts/user logs in.
  - Can be changed for users, computers, and domain controllers.

GPMC Container Tabs

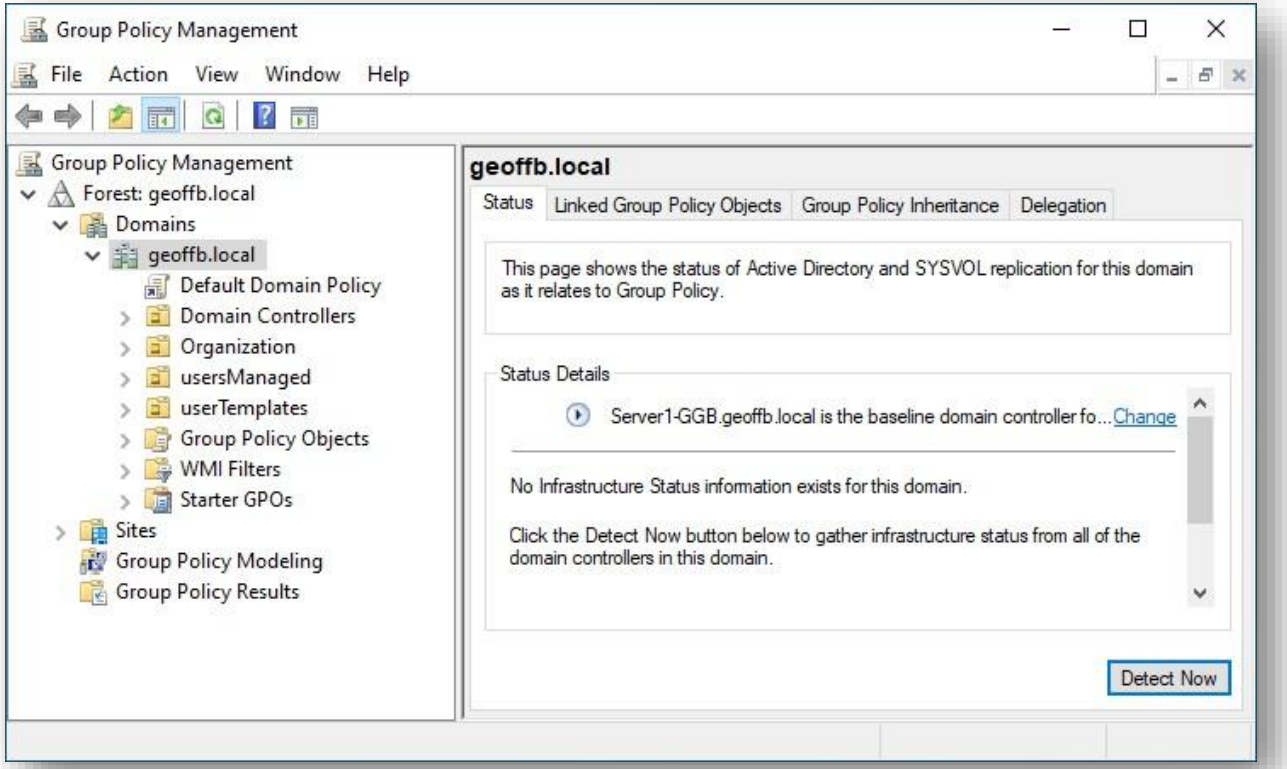
Group Policy Management Console

(GPMC; GPMC.msc)



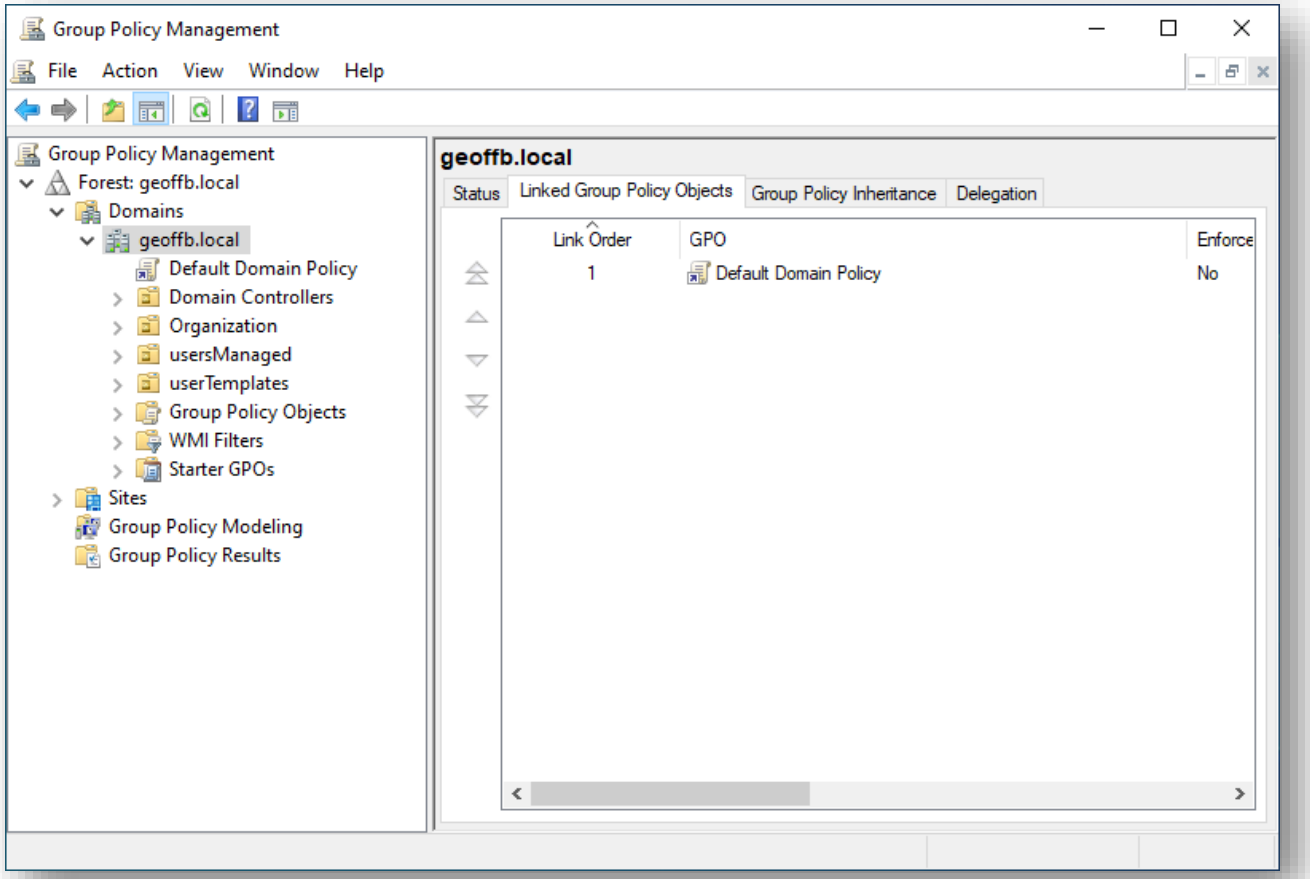
- Every container in GPMC (site, domain or OU) has three tabs in the right pane:
  - Linked Group Policy Objects – GPOs linked to the site, domain or OU
  - Group Policy Inheritance – order of precedence for GPOs applied
  - Delegation – who has permissions to the GPO
  - Note: Status – the status tab only shows on the domain container.

GPMC: Container Tabs: Linked GPOs



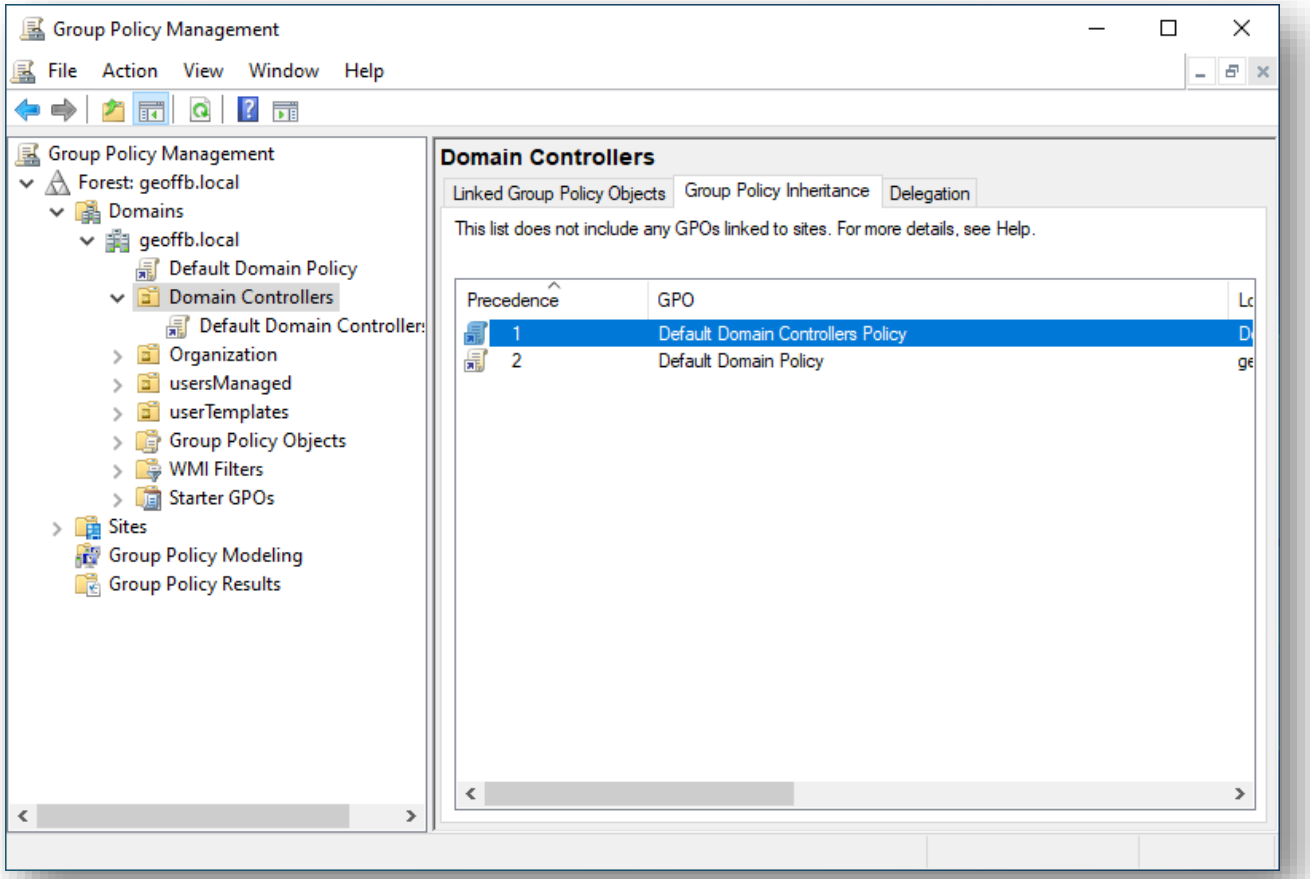
- Allows you to test you replication of GPOs between domains in the forest.

GPMC: Container Tabs: Linked GPOs



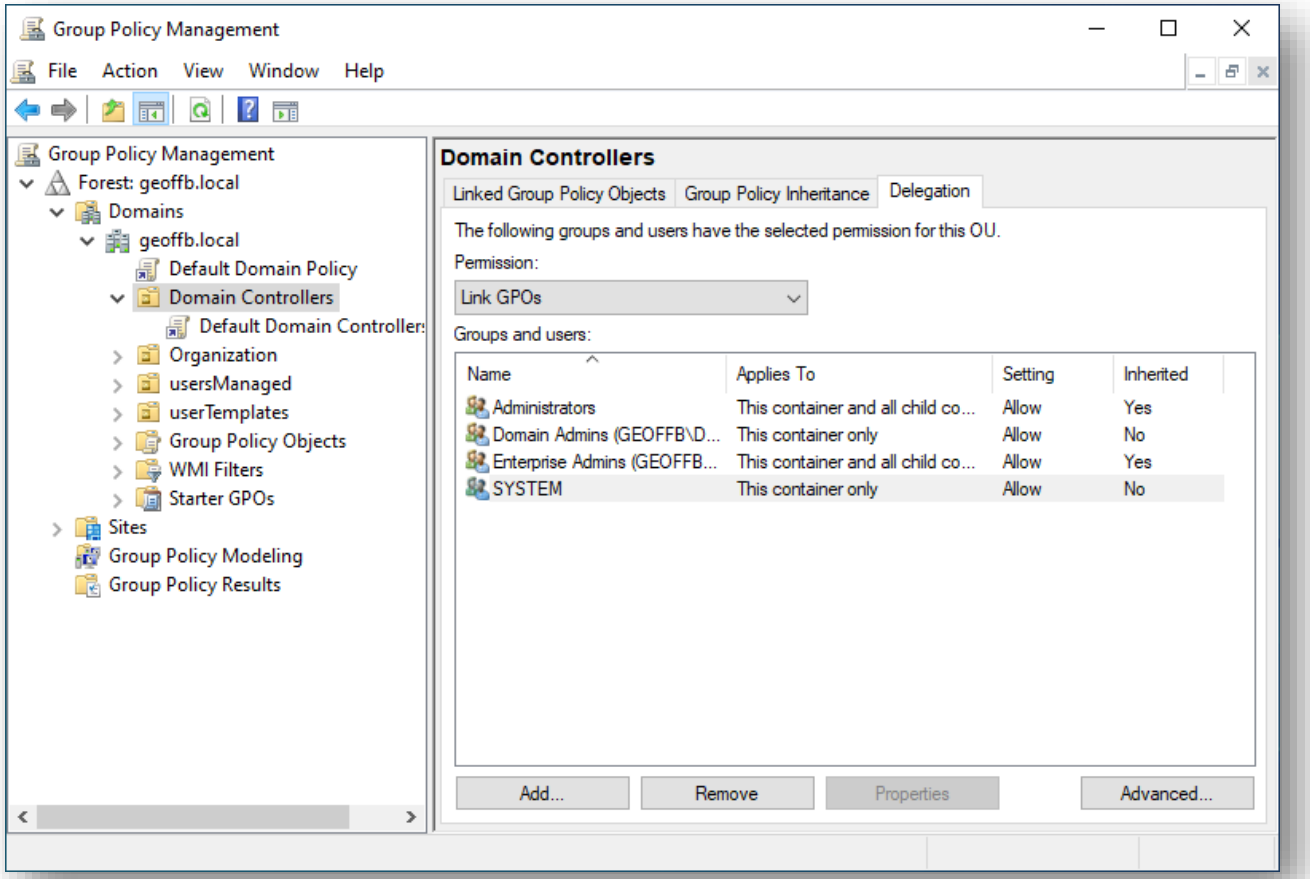
- This tabs shows you which GPOs are linked to this container and the link order of those GPOs.
- GPO enforcement is also shown here.

GPMC: Container Tabs: GP inheritance



- This tab shows all policies linked to this object because they are inherited from a parent object.

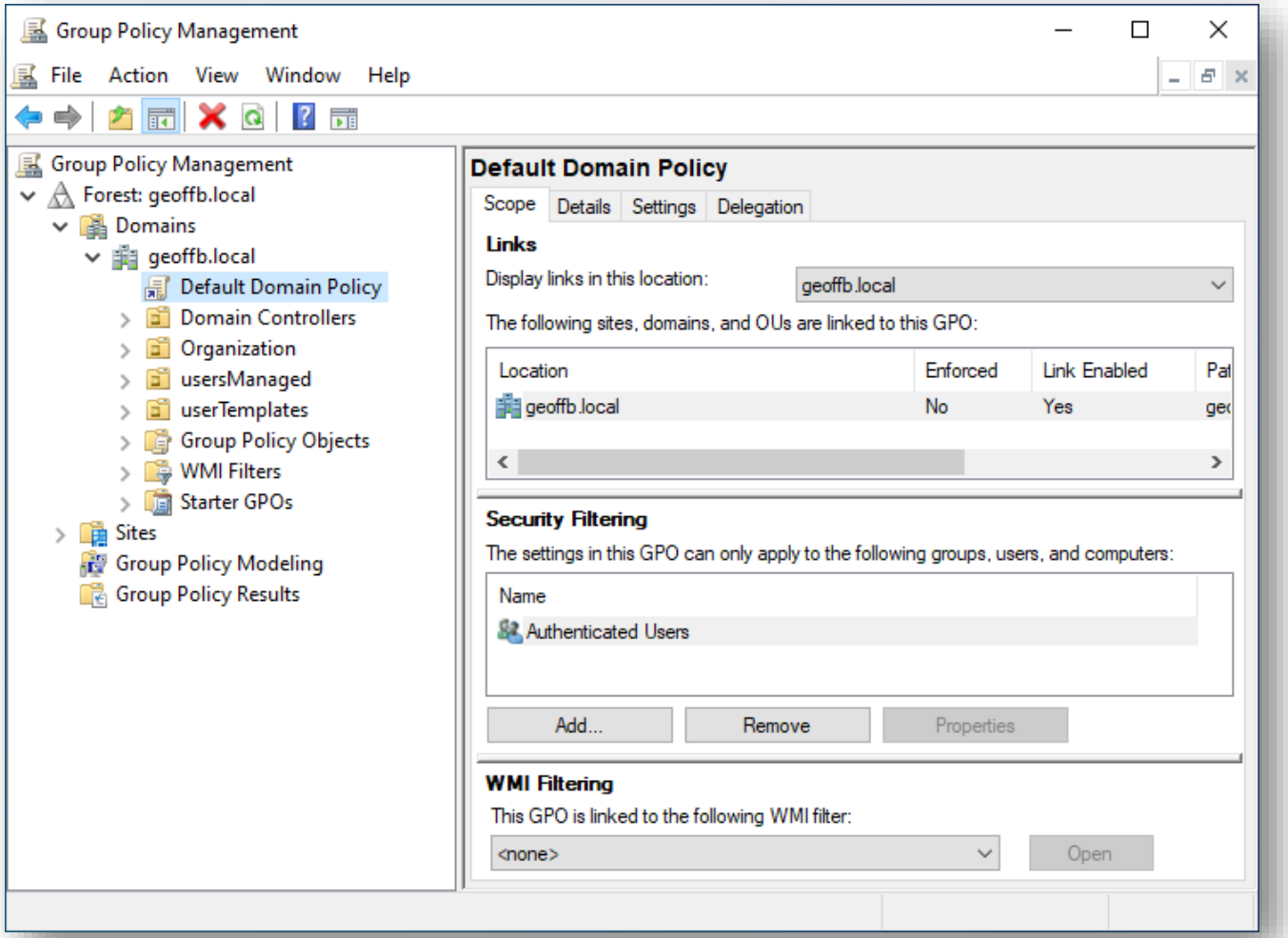
GPMC: Container Tabs: Delegation



- This tab allows you to delegate GPO-related container permissions to non-administrative accounts.
- Use the Advanced button to see the permissions sheet.

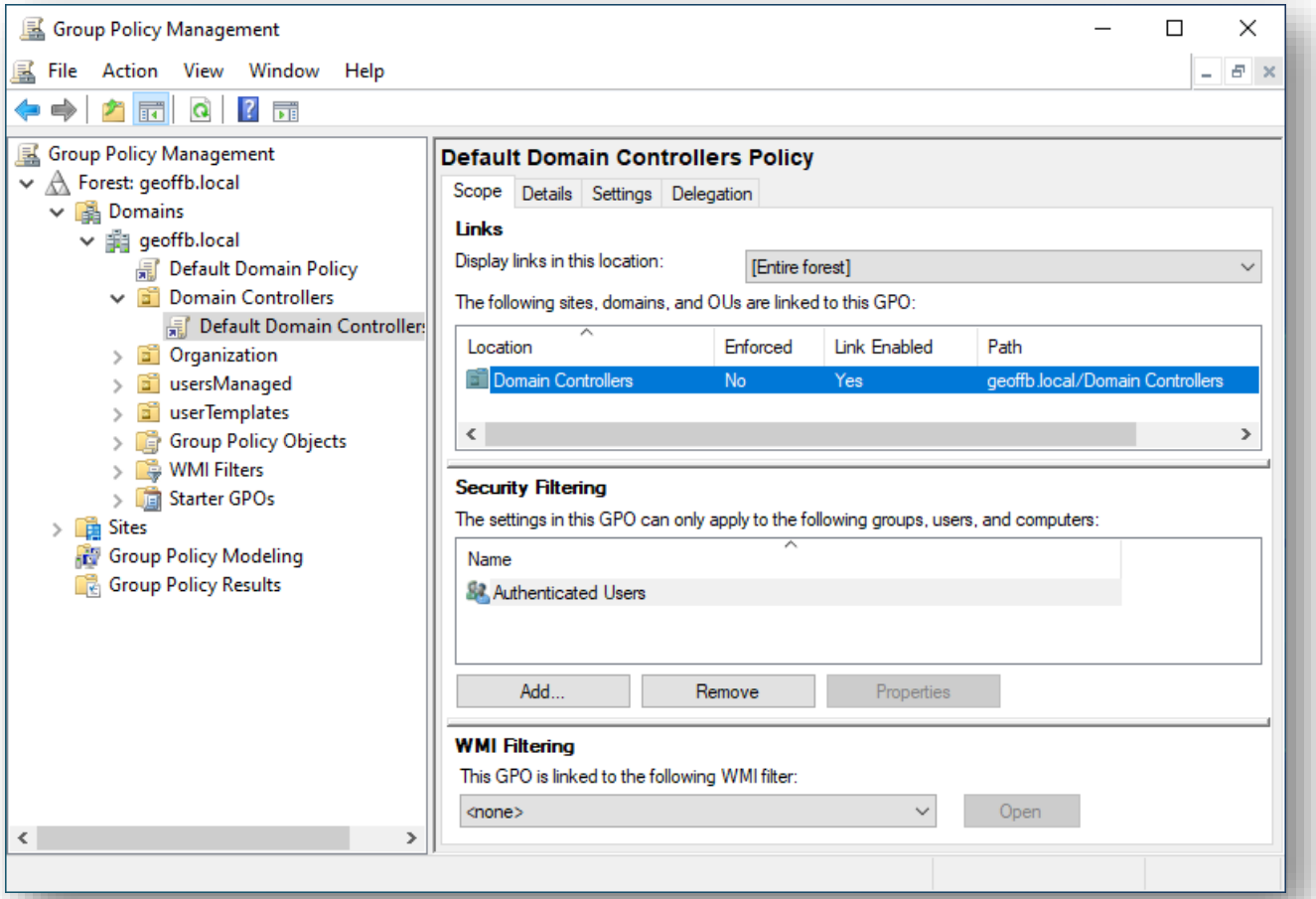
GPMC GPO Tabs

Group Policy Objects



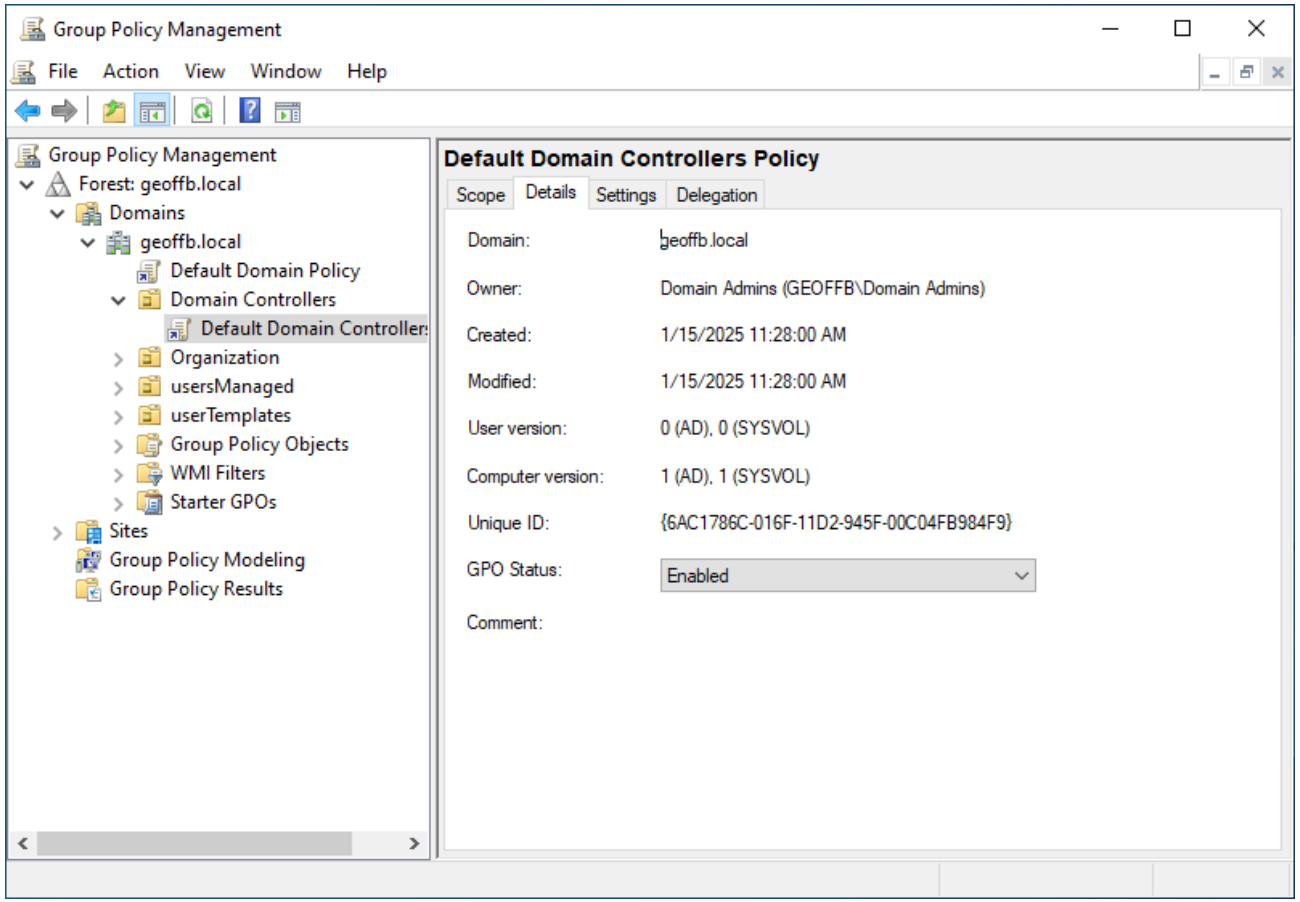
- Each GPO, when selected, has four tabs:
  - Scope – which containers policy is linked, who the GPO applies to; what WMI filters are in place
  - Details – Enable or disable GPO, displays read-only information about the GPO
  - Settings –displays GPO setting areas that have values other than the original default values.
  - Delegation – Lists the users/groups that have permissions to this GPO

GPMC: GPO Tabs: Scope



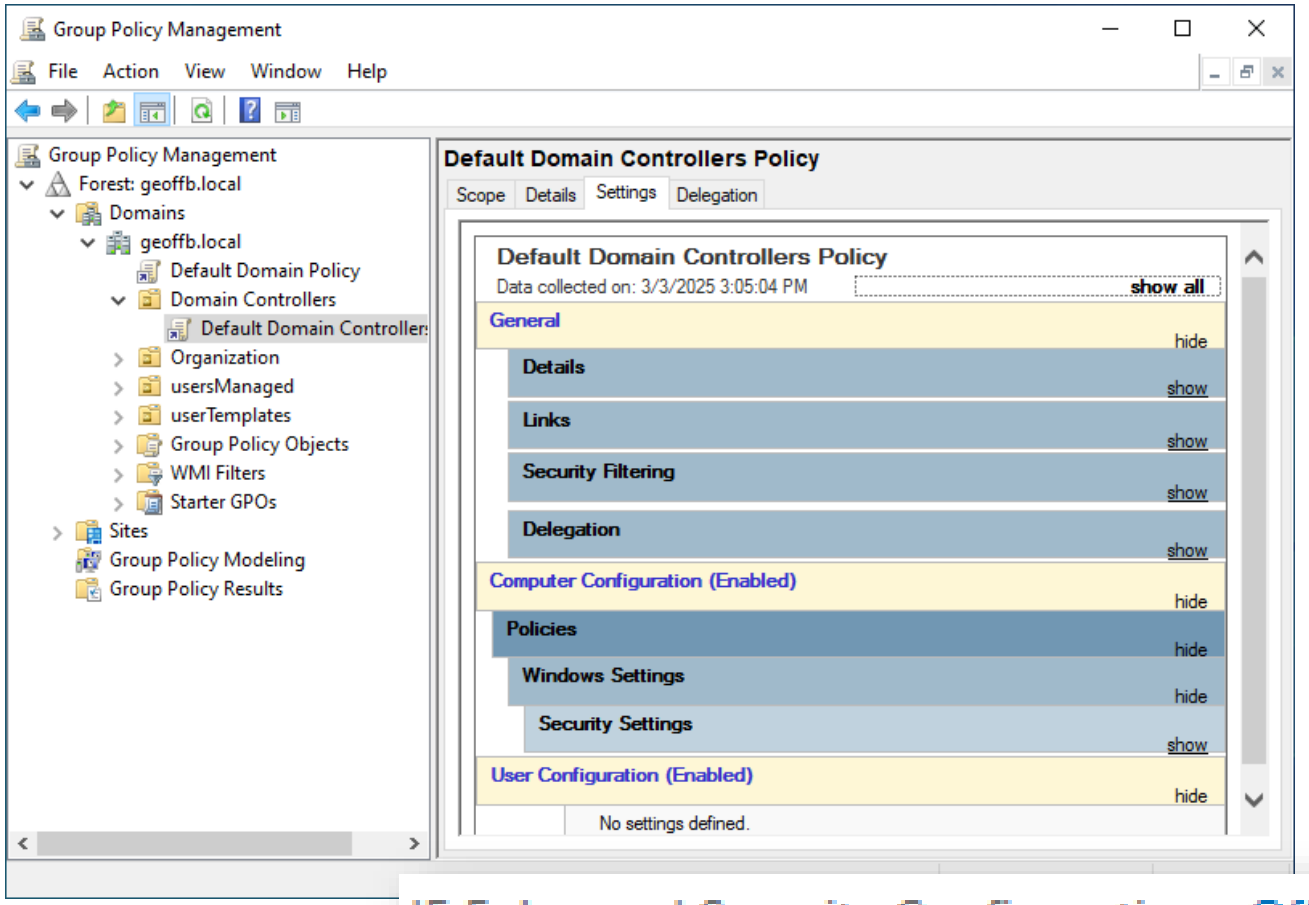
- Links allows you to see where the GPO is applied (GPOs can be applied to *more than one container*).
- Security Filtering shows who can read and apply the settings in the GPO.
- WMI filters can be created based on the target computer's make, model, operating system, time zone, etc.

GPMC: GPO Tabs: Details



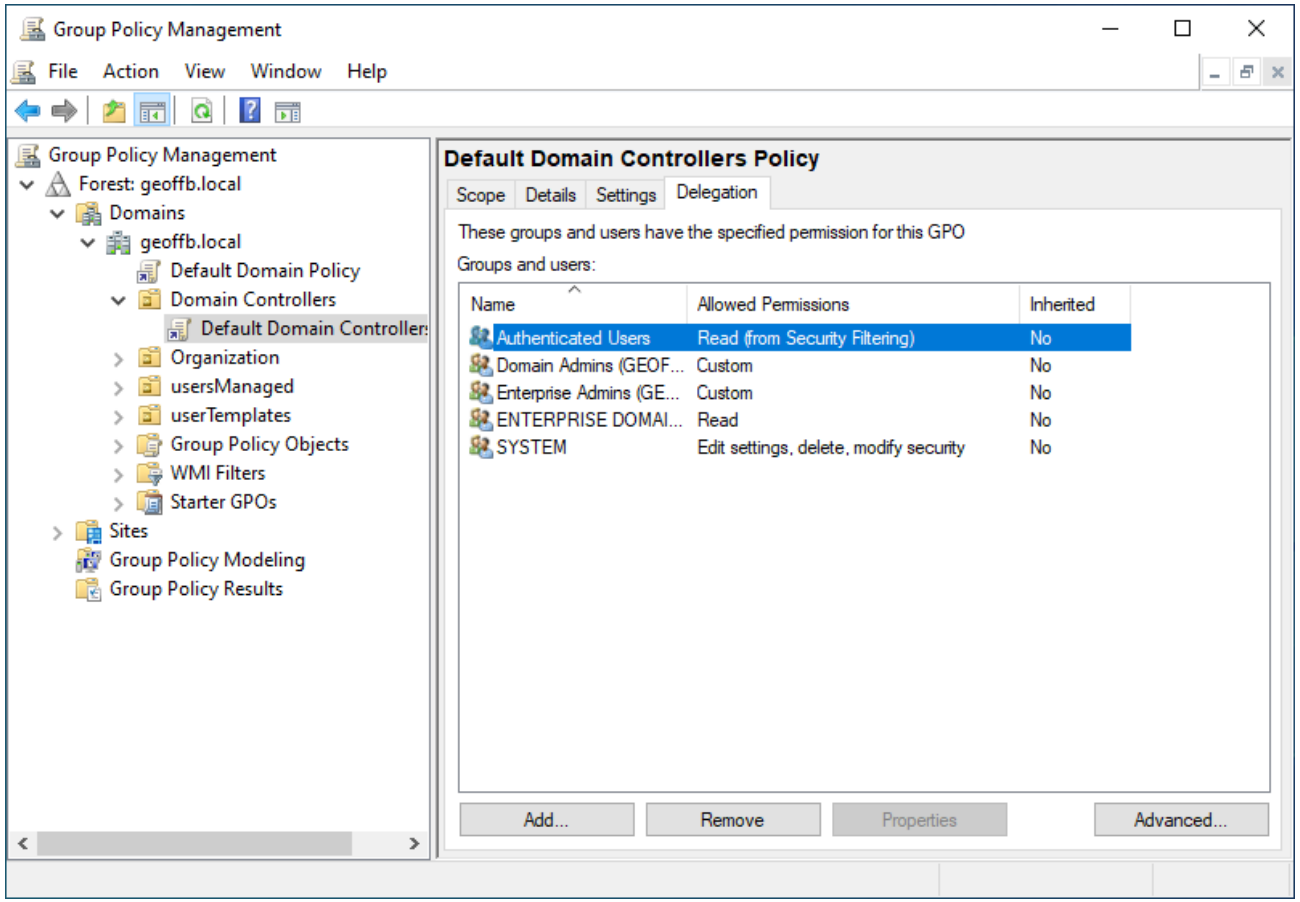
- In details we can see information about the GPO as an object.
  - Time created and modified
  - The domain the GPO exists in
  - The GUID (unique ID) of the GPO.
  - Whether the GPO is enabled/disabled.
  - Any additional comments.

GPMC: GPO Tabs: Settings



- Any setting that is configured will be shown here. This is the best place to go to quickly see the settings configured in a GPO.
- Only configured settings appear here.
- Must turn off IEESC for this to work (server manager).

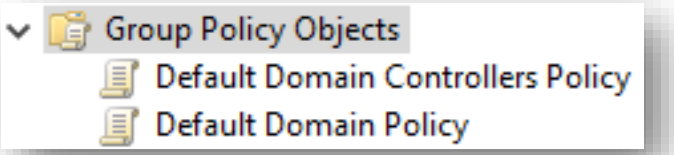
GPMC: GPO Tabs: Details



- This tab allows you to delegate GPO-related permissions to non-administrative accounts.
- Can also be used to *limit* permissions to read or modify GPOs.s
- Use the Advanced button to see the permissions sheet.

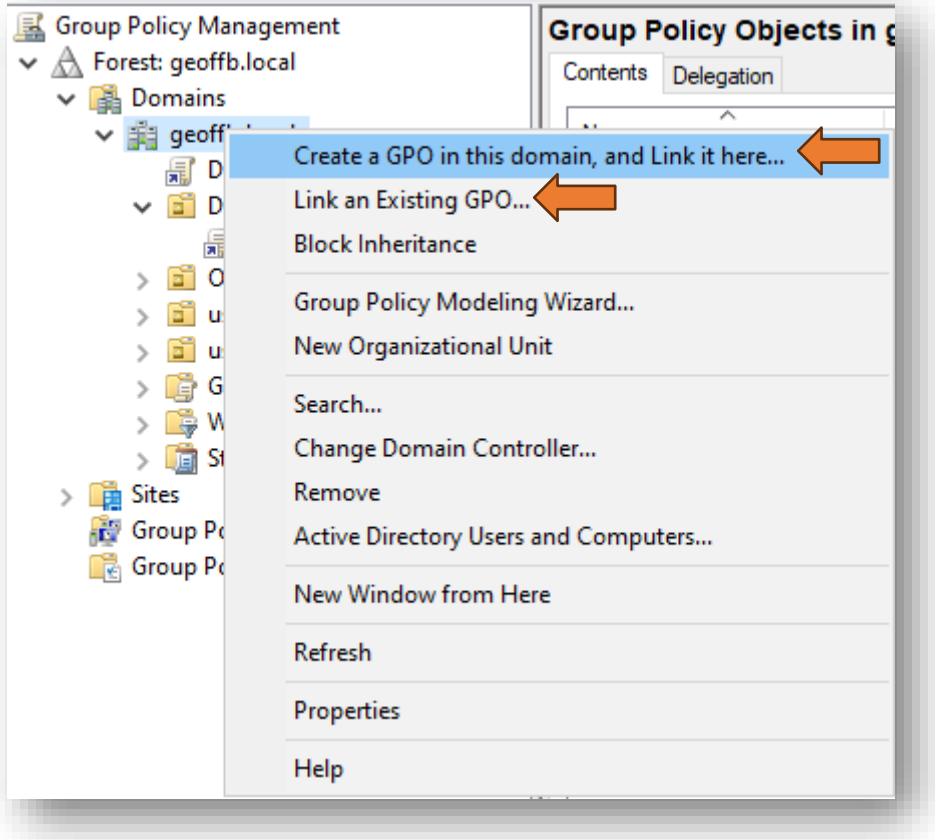
Creating vs Linking GPOs

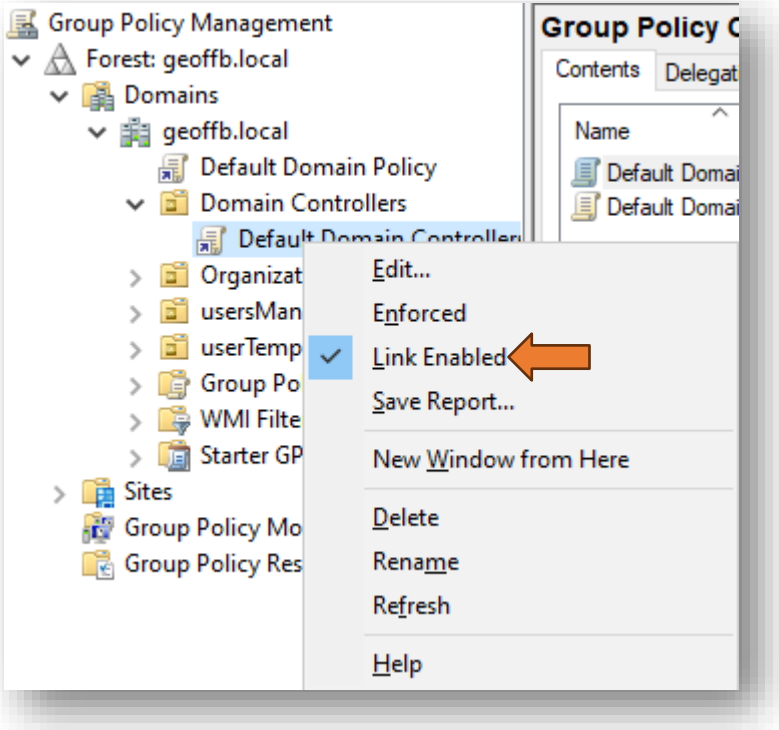
Creating vs. Linking



- When a user creates a new GPO is stored in the Group Policy Objects folder. All GPOs are stored here.
- When a user creates a new Group Policy Object (GPO), they must create a new link to the target OU where the GPO is to be applied.
  - (A link essentially "applies" that GPO to that container. A single GPO can be applied to many AD containers simultaneously)
- For a policy to apply to any user or computer, the policy must be linked to the domain, a site, or an OU where the object resides.
  - (Inheritance **does** apply for child containers as well)
- When you right-click a location and choose Create a GPO in this domain and link it here you are performing both actions at the same time (creating and linking)
- You can disable a link without removing the GPO from the OU.
- Where you remove a link, the GPO is NOT deleted, only the link is.

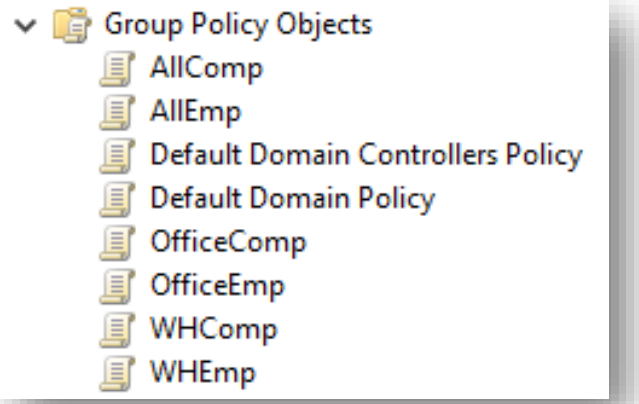
Creating and Linking

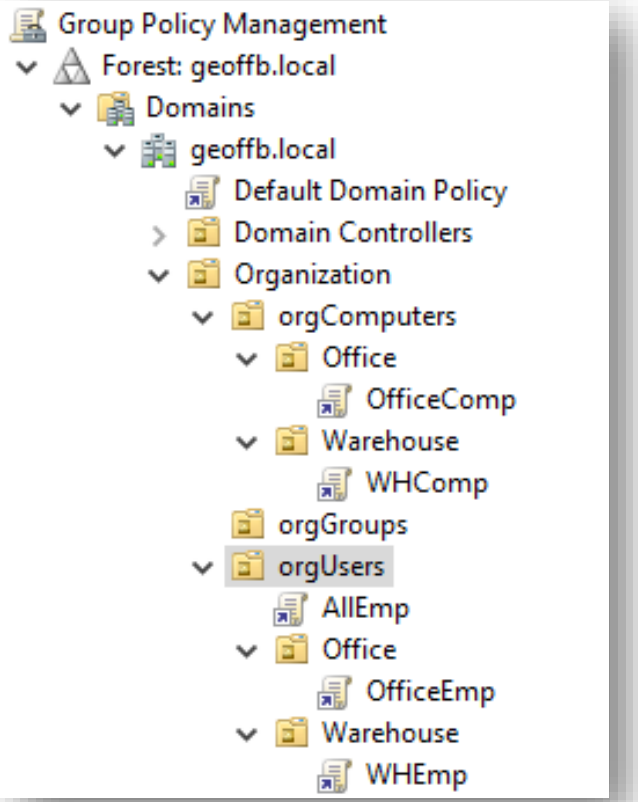




Do this…

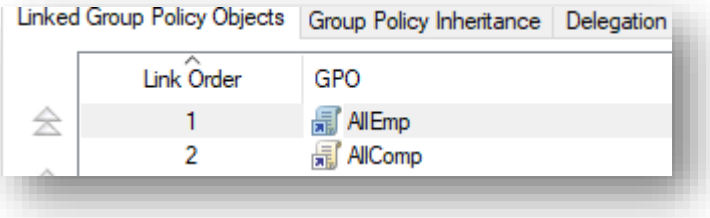
1. In GPMC create a GPO called AllEmp in the Group Policy Objects Container
2. In GPMC create a GPO called AllComp in the Group Policy Objects Container
3. Create and link a GPO called OfficeEmp in the orgUsers/Office OU
4. Create and link a GPO called WHEmp in the orgUsers/Warehouse OU
5. Create and link a GPO called OfficeComp to orgComputers/Office
6. Create and link a GPO called WHComp to orgComputers/Warehouse
7. Link the AllEmp GPO to the orgUsers OU



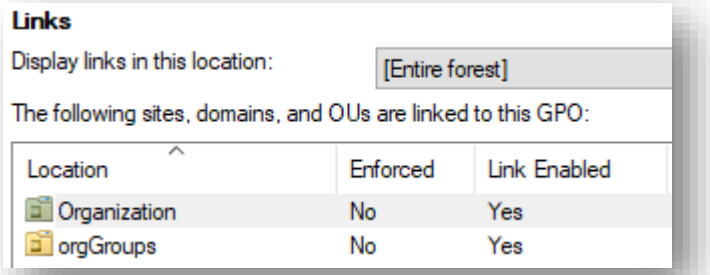


Do this…

- Look at each container with a linked GPO.
  - Linked GPOs: shows any GPO explicitly linked here. Link order is set by the order in which the GPOs were initially linked.
  - GP Inheritance: GPOs linked to my parent objects.



- Look at the GPO linked to a container
  - Scope: notice where the GPO is linked. If the GPO is linked in more than one place, each container will be shown here.



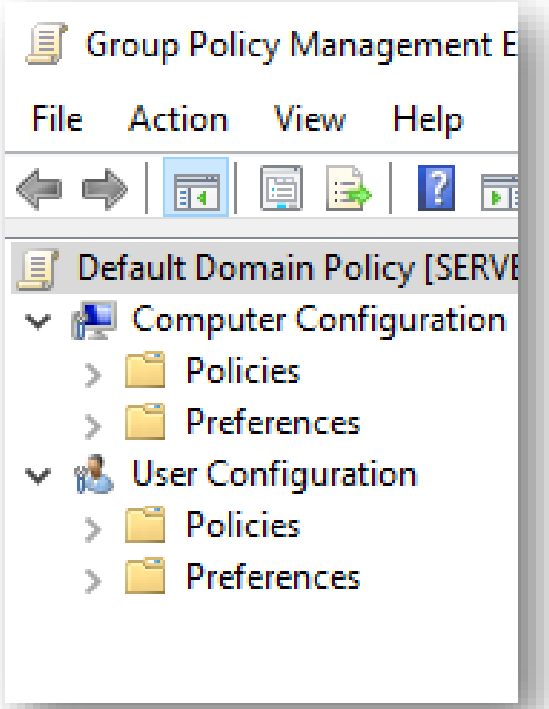
Understanding Group Policy Concepts

Start Day 2

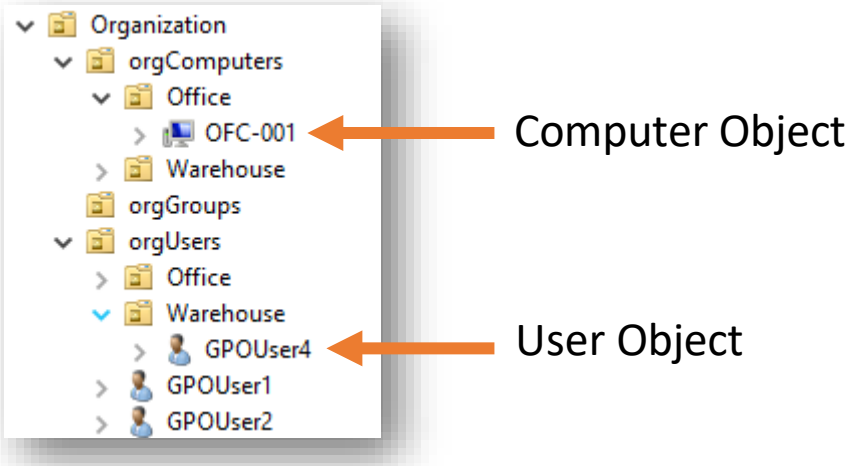
Understanding Group Policy Concepts

- Each GPO contains every possible setting. ONLY the settings that are configured will affect the object the GPO is applied to.
- When applied, the settings in the GPO are written to registry.
  - Computer settings are written to HKLM when the computer authenticates to AD (at boot).
  - User settings are written to HKCU after the user authenticates successfully.
- GPOs can affect computers or users or both.
- GPOs can be linked to one or more OUs.

Understanding Group Policy Concepts

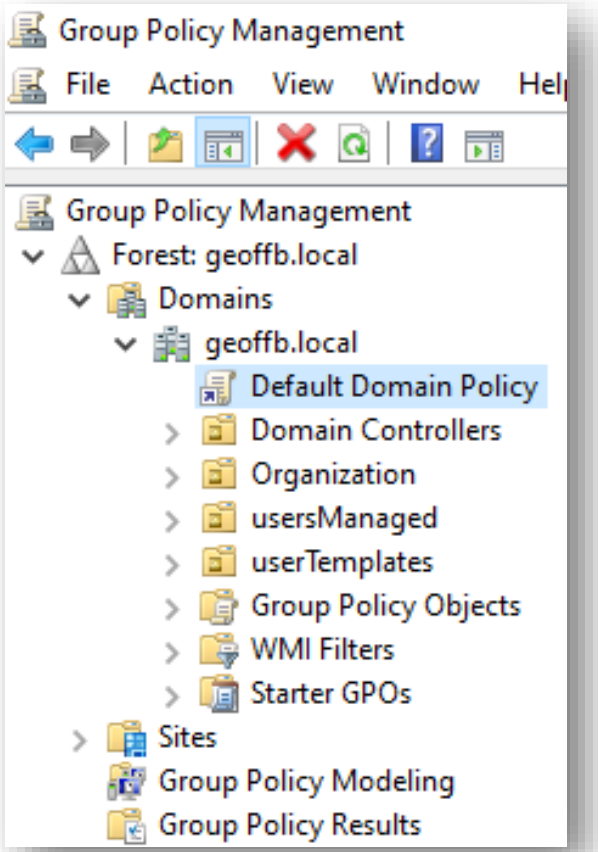


- Computer configuration settings apply only to computer objects within the container the GPO was applied to.
- User configuration settings of a GPO apply only to user object within the container the GPO was applied to.



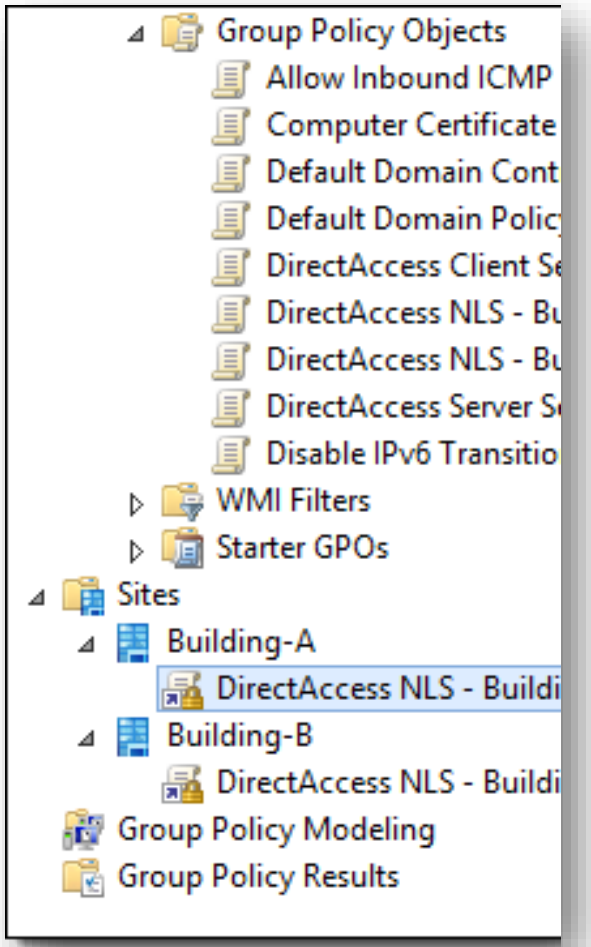
ADUC

Understanding Group Policy Concepts



- A GPO linked to a site, domain or OU applies to the user and computer objects that exist in the container (and child containers) to which they are linked.
- In this example, the Default Domain Policy, GPO - applied to the root of the Domain (the domain container) - applies to *all* computer objects and *all* user objects in the domain (due to the parent/child relationship of the containers).

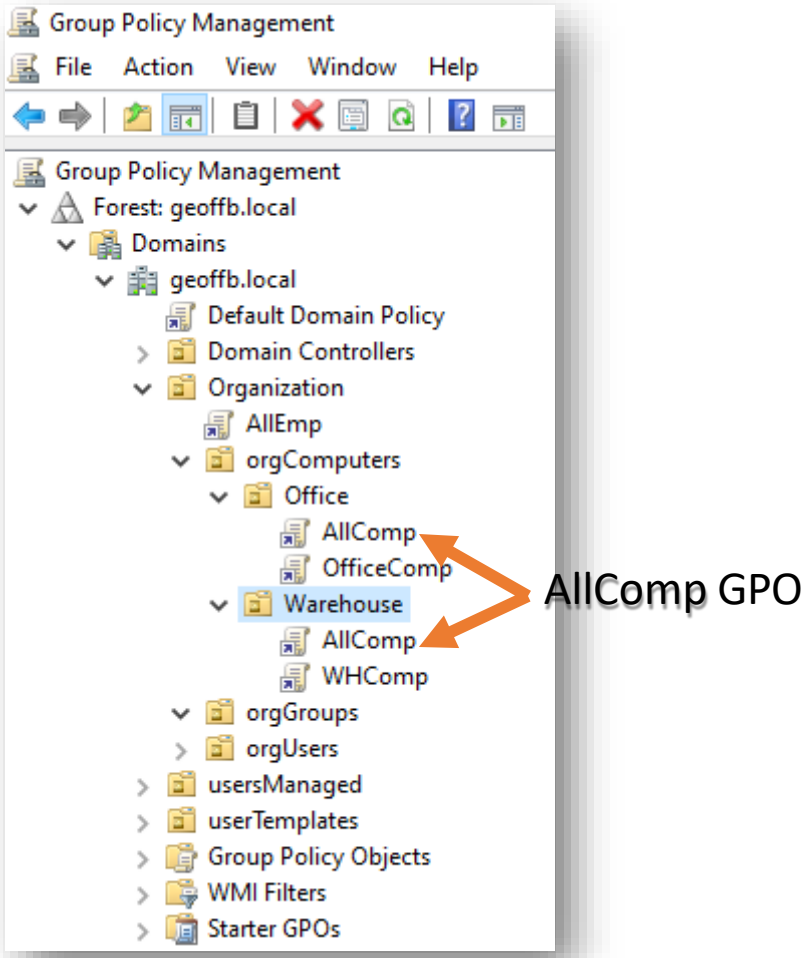
Understanding Group Policy Concepts



- A GPO applied to a site applies to all the computer objects that physically exist within the site and to users who logon to computers in that site, regardless of their domains or OUs

Understanding Group Policy Concepts

- A single GPO can be linked to multiple OUs, sites or even domains. However, they are always stored on the DCs of a domain, in the domain where they were created.



Do this…

- Link the AllComp GPO to the orgComputers/Office and orgComputers/Warehouse OUs.

  Consider: this policy would apply to all of our computers in Organization, but how is this different from how we applied AllEmp?

Understanding Group Policy Concepts

- Group policy settings do not take effect immediately unless the GPO is a Local GPO.
- How/when does GPO apply to a computer?
  - Computer settings are written to HKLM when the computer authenticates to AD (at boot).
  - User settings are written to HKCU after the user authenticates successfully.
  - Approximately 90 minutes has passed since the last update cycle.
  - GPO is applied manually VIA the CLI.

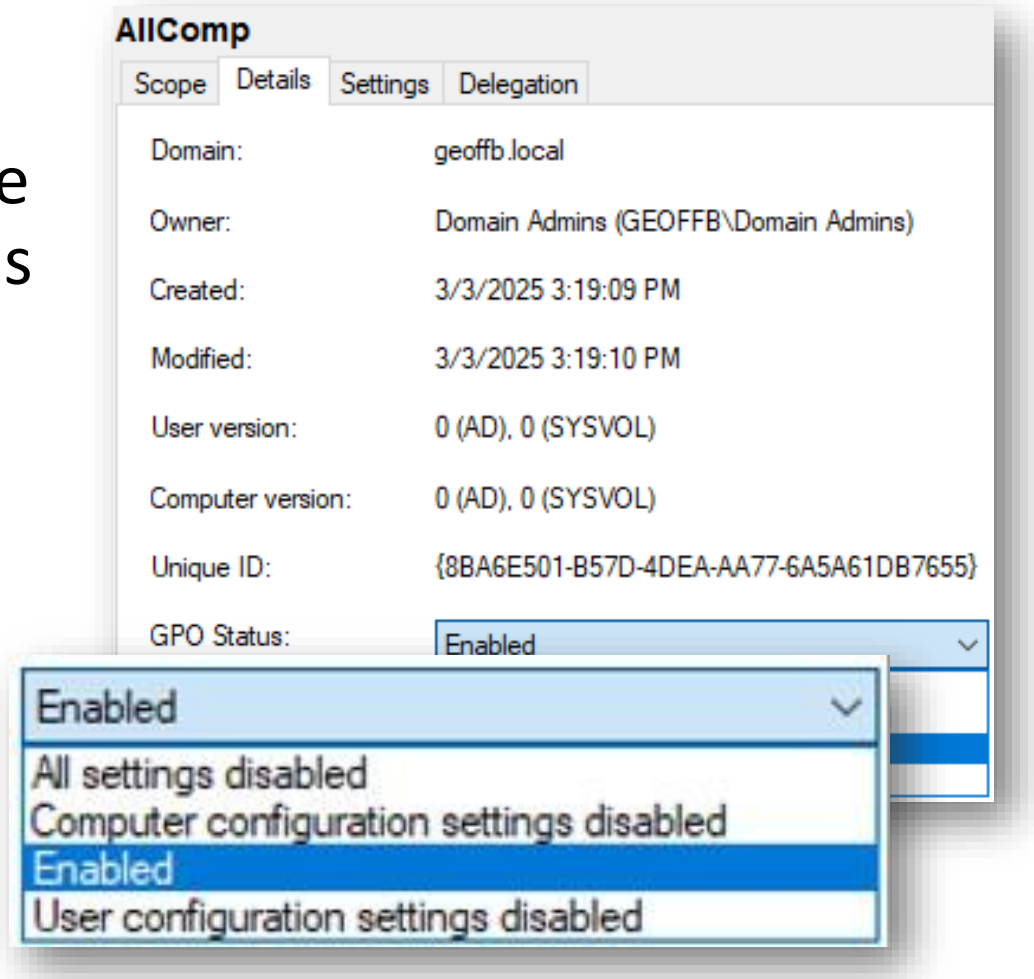
Understanding Group Policy Concepts

- To apply a GPO manually (to *force* an update) you must run the:
  - `gpupdate` (cmd)

    OR
  - `Invoke-GPUpdate` (ps)
  - `/force` (cmd) or `-force` (ps) will update *all* settings applicable in all GPOs applied to that object. By default, only the police settings that are new or have changed will be updated.

Disabling Unused Settings in a GPO

- If a GPO contains settings in only one of the configuration nodes of the group policy (i.e. all policy settings are in the User Configuration section), it is a best practice to disable the unused node.
- Disabling unused settings
  - Select a GPO and go to its details tab
  - Under GPO status, select the appropriate setting



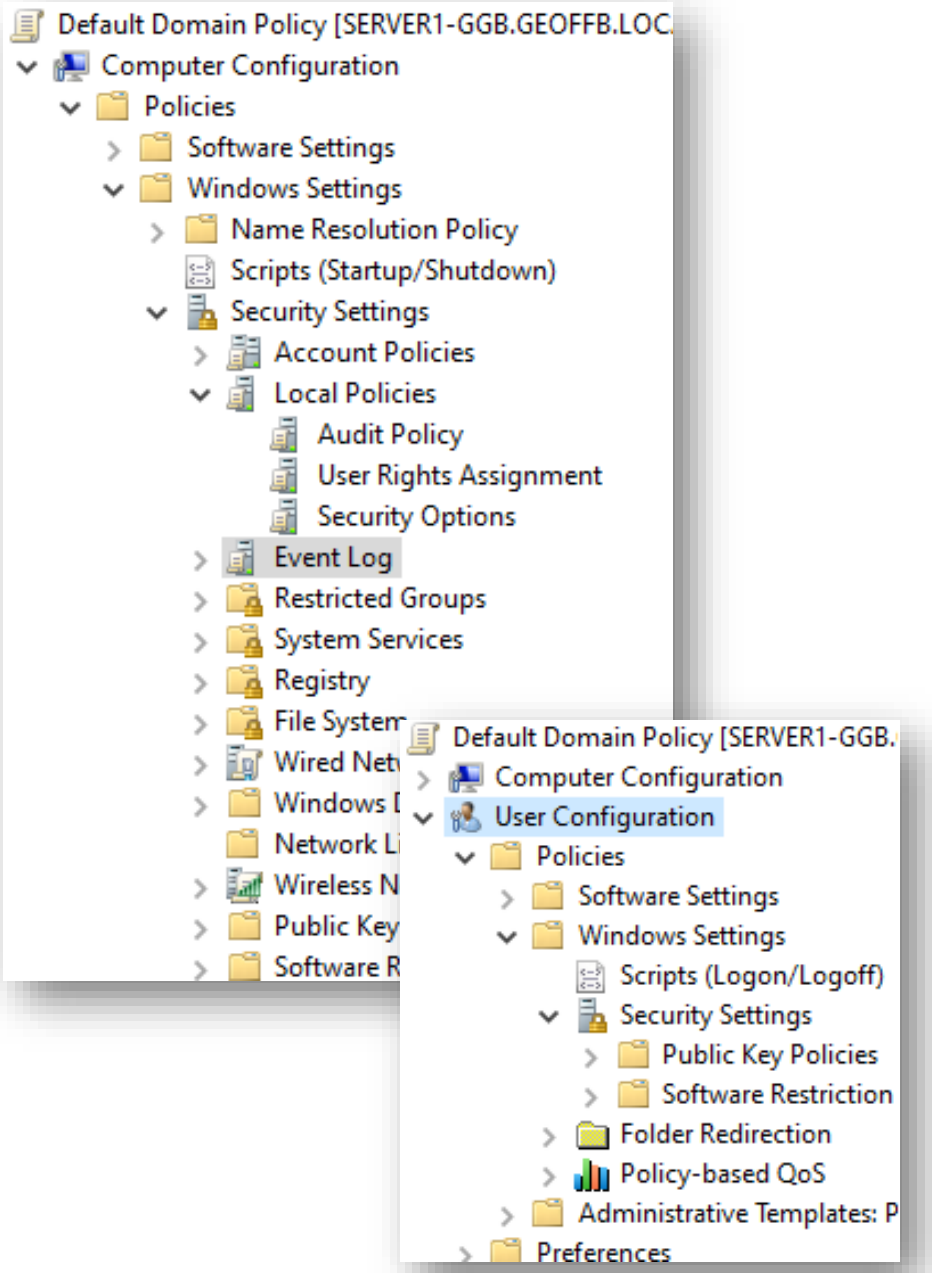
Do this…

1. On ALLComp GPO, disable the User Policies
2. On ALLEmp GPO, disable the Computer Policies

GPO Settings

Security Settings

- Mostly found only in computer settings.
- Used to secure the environment.
- Settings can *typically* be ON, OFF, or NOT DEFINED. How the setting is configured varies depending on the policy setting
- Examples:
  - Password/lockout polices
  - User Rights Assignments
  - Environment Security Options
  - Auditing
  - Restricting Group Local group membership
  - Controlling System Services
  - Firewall configuration

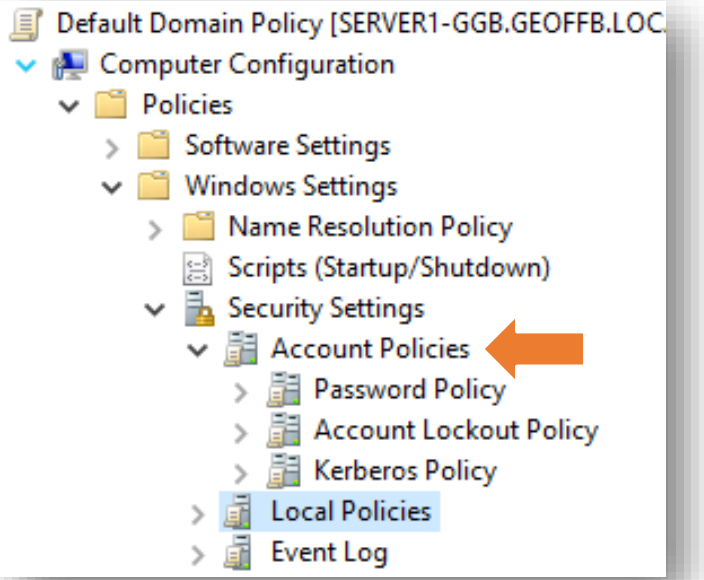


User Settings

And much more…

Account Policies

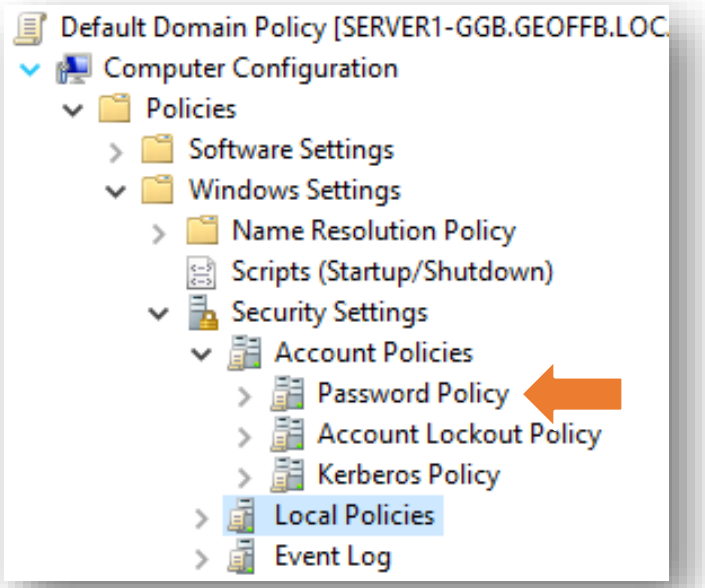
- Security measures for a computer.
- Account policies focus on:
  - Password security
  - Account lockout
  - Kerberos security



Account policies can **only** be applied at the domain or local computer levels, account policies applied at the OU or Site level *will have no effect*. This means you can have **one password policy per domain.**

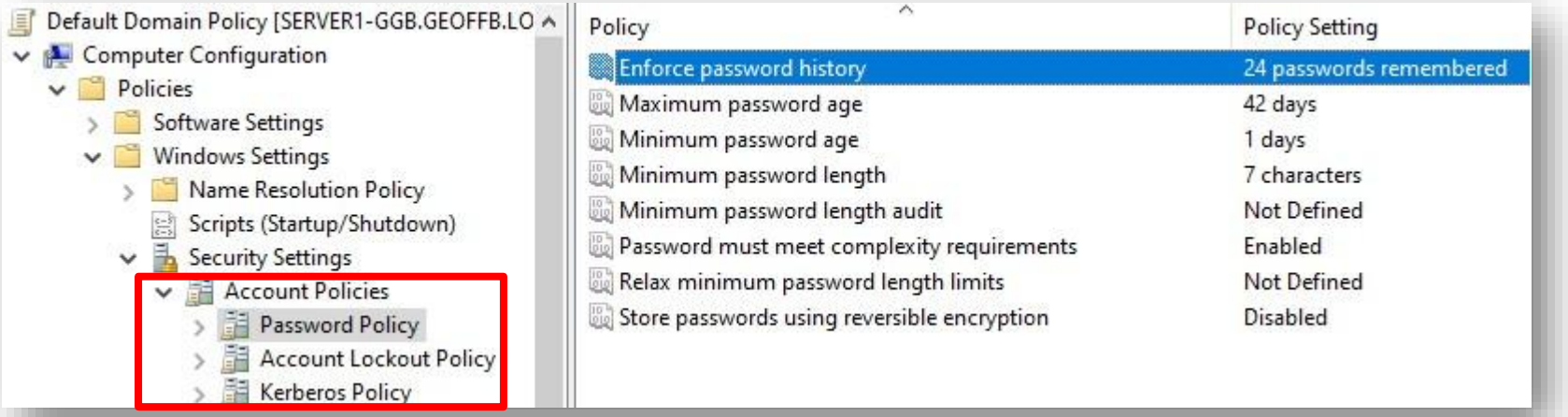
Password Policies

- Password policies configure the requirements for user passwords in the domain.
- They are found in: *Computer Configuration\Windows Settings\Security Settings\Account Policies\Password Policies*
- These policy settings can only be applied on the Domain Container; applying them to sites or OUs will have no effect.



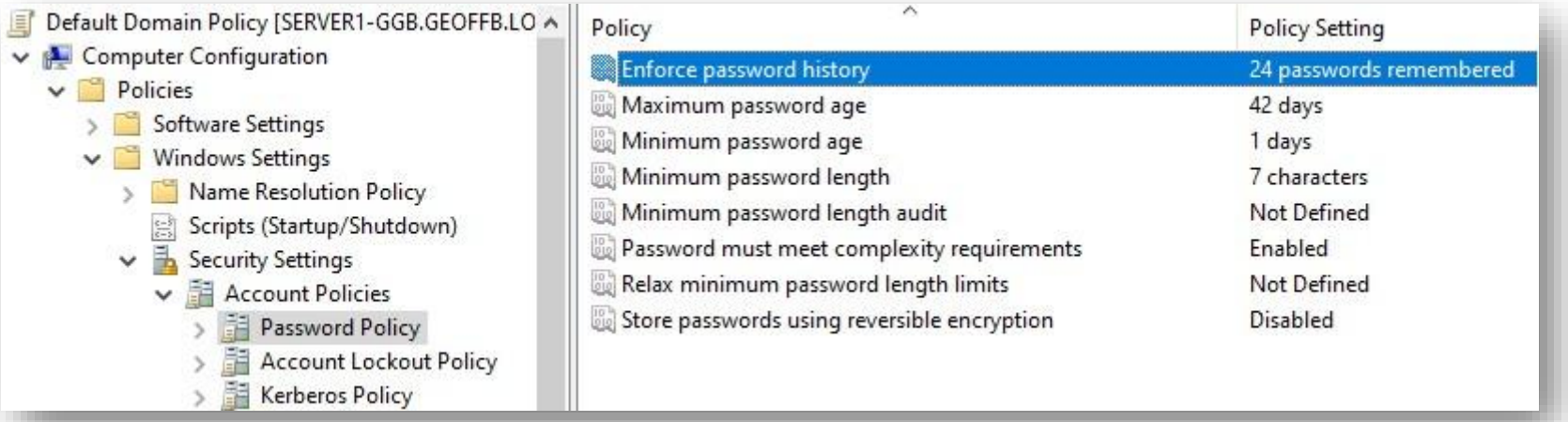
Password Policies

- *Computer Configuration\Windows Settings\Security Settings\Account Policies\Password Policies*
- Enforce password history
  - Enables you to require users to choose new passwords when they make a password change, because the system can remember the previously used passwords
  - Default value in a domain is 24 passwords remembered (set by the Default Domain policy.)



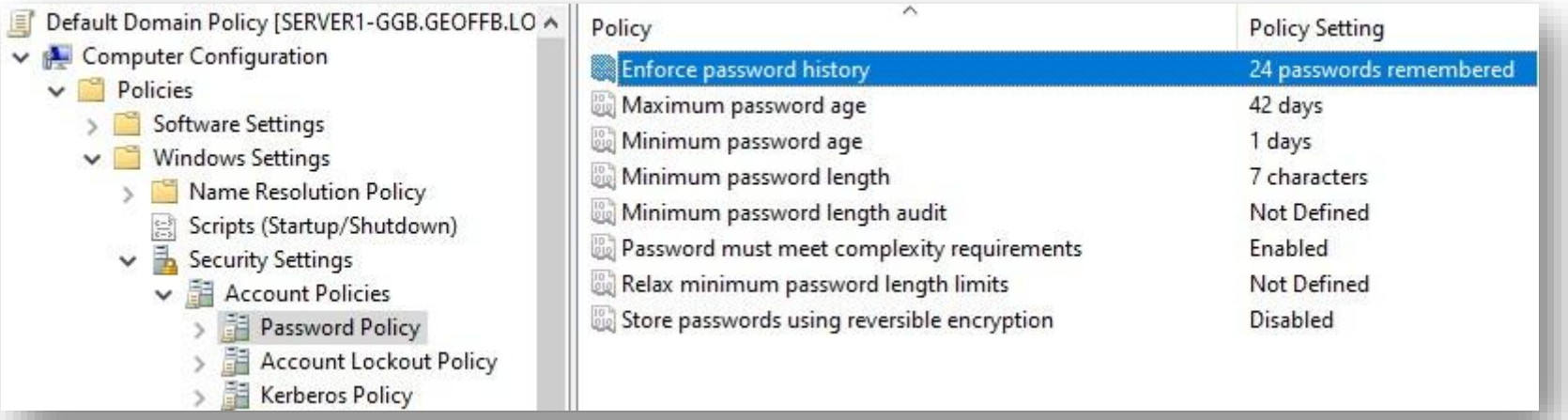
Password Policies

- Maximum password age
  - Permits you to set the maximum time allowed until a password expires
  - Default value is 42 days
- Minimum password age
  - Permits you to specify that a password must be used a minimum amount of time before it can be changed
  - Default value is 1 day



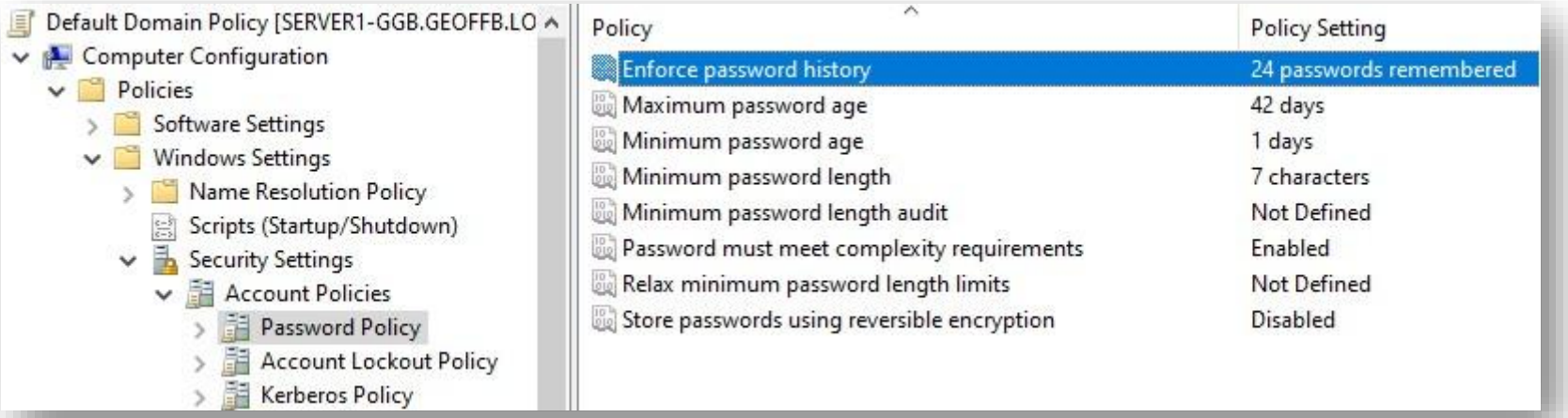
Password Policies

- Minimum password length
  - Enables you to require that passwords are a minimum length
  - Default value is eight characters
- Minimum password length audit
  - The minimum password length for which password length audit warning events are issued



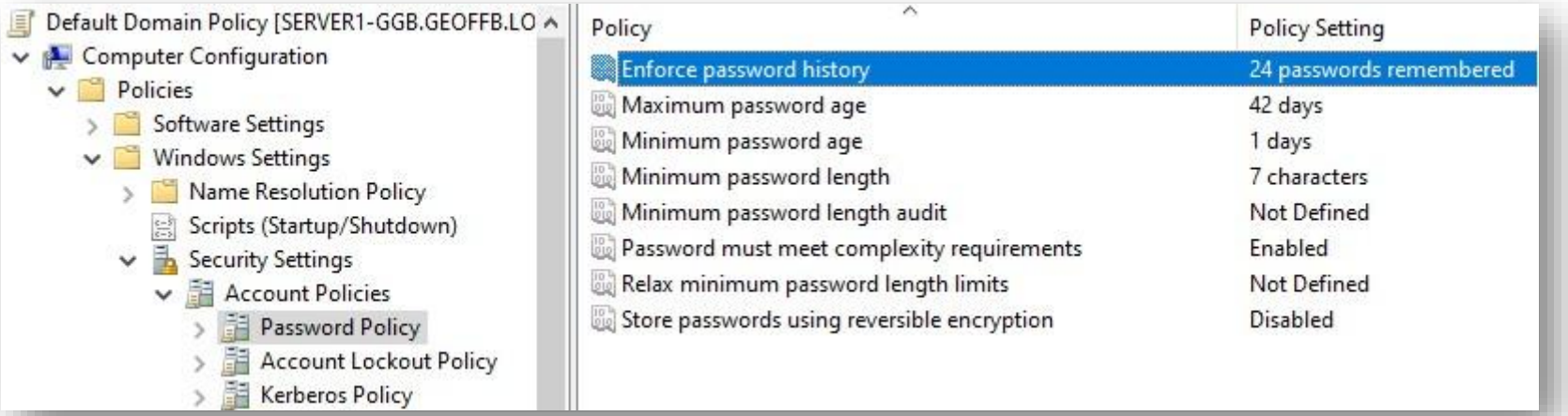
Password Policies

- Passwords must meet complexity requirements:
  - Enables you to create a filter of password requirements that each account password must follow
  - Criteria include at least six characters; no duplication of user's account name; inclusion of characters from at least three of the following four categories: uppercase letters, lowercase letters, numbers and symbols.
  - Policy is enabled by default



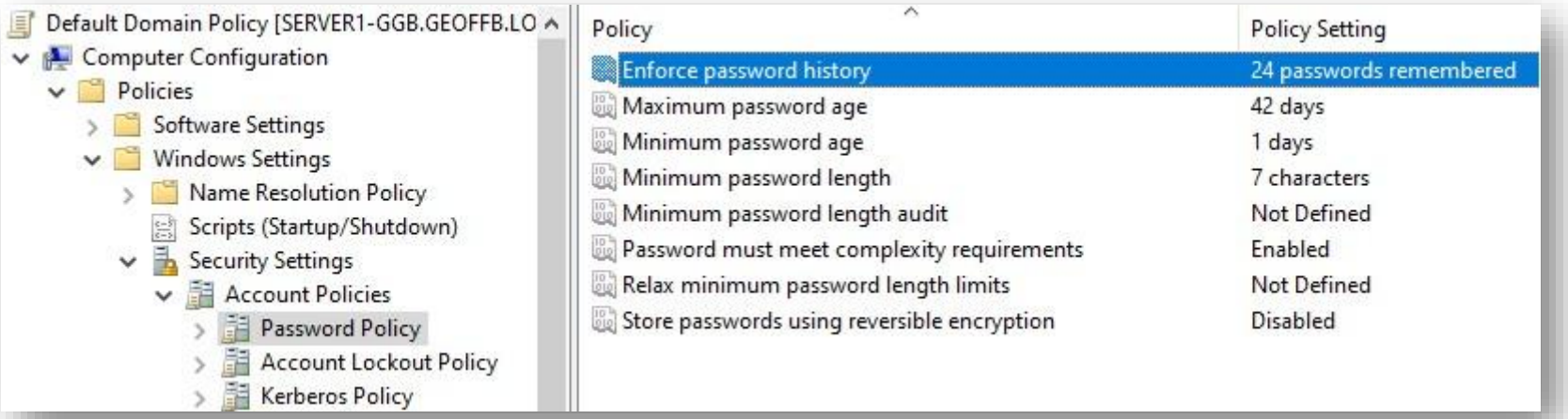
Password Policies

- Relax minimum password length limits
  - This setting controls whether the minimum password length setting can be increased beyond the legacy limit of 14.
  - Default is “not defined” by default



Password Policies

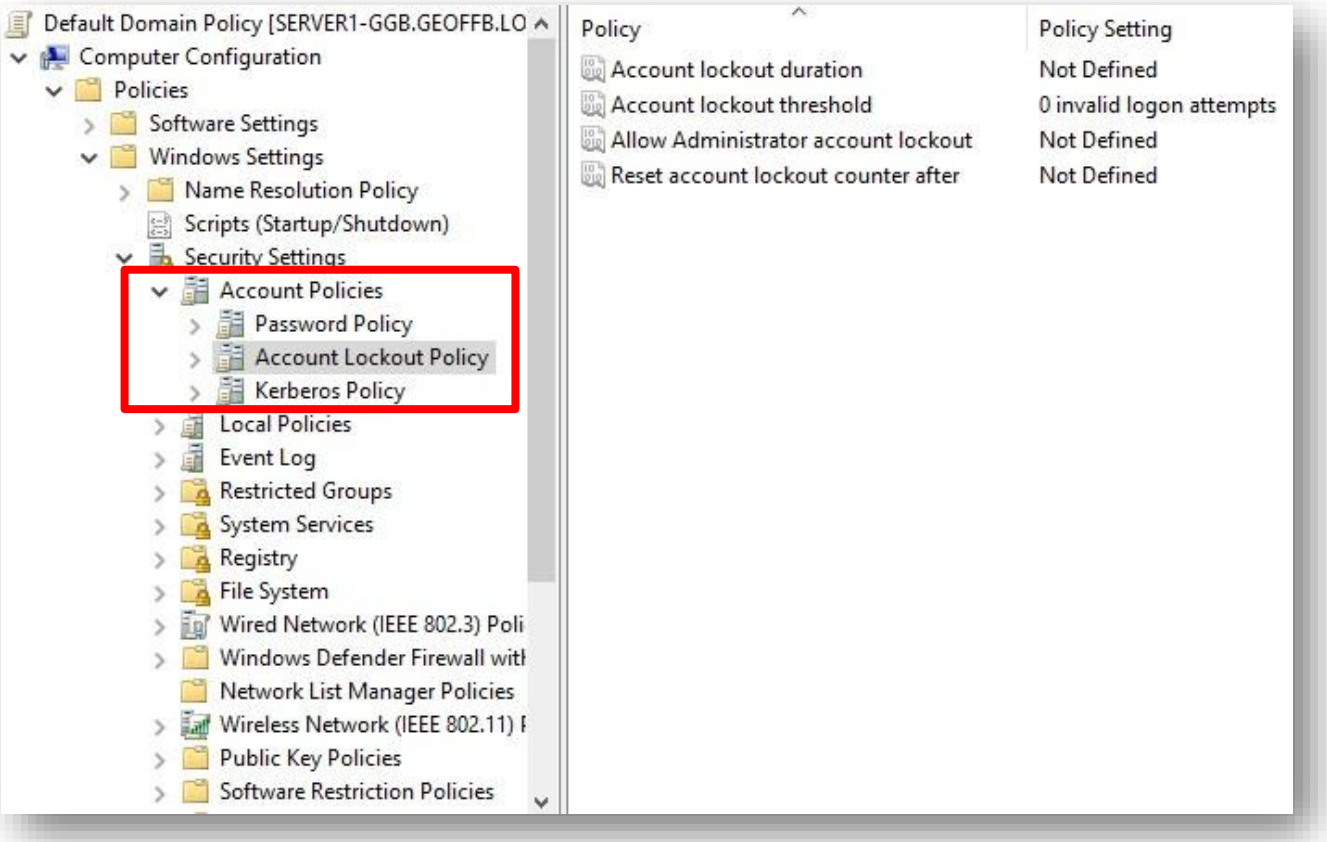
- Store passwords using reversible encryption
  - This security setting determines whether the operating system stores passwords using reversible encryption.
  - This policy provides support for applications that use protocols that require knowledge of the user's password for authentication purposes.
  - This policy should never be enabled unless application requirements outweighs the need to protect password information.



Account Lockout Policy

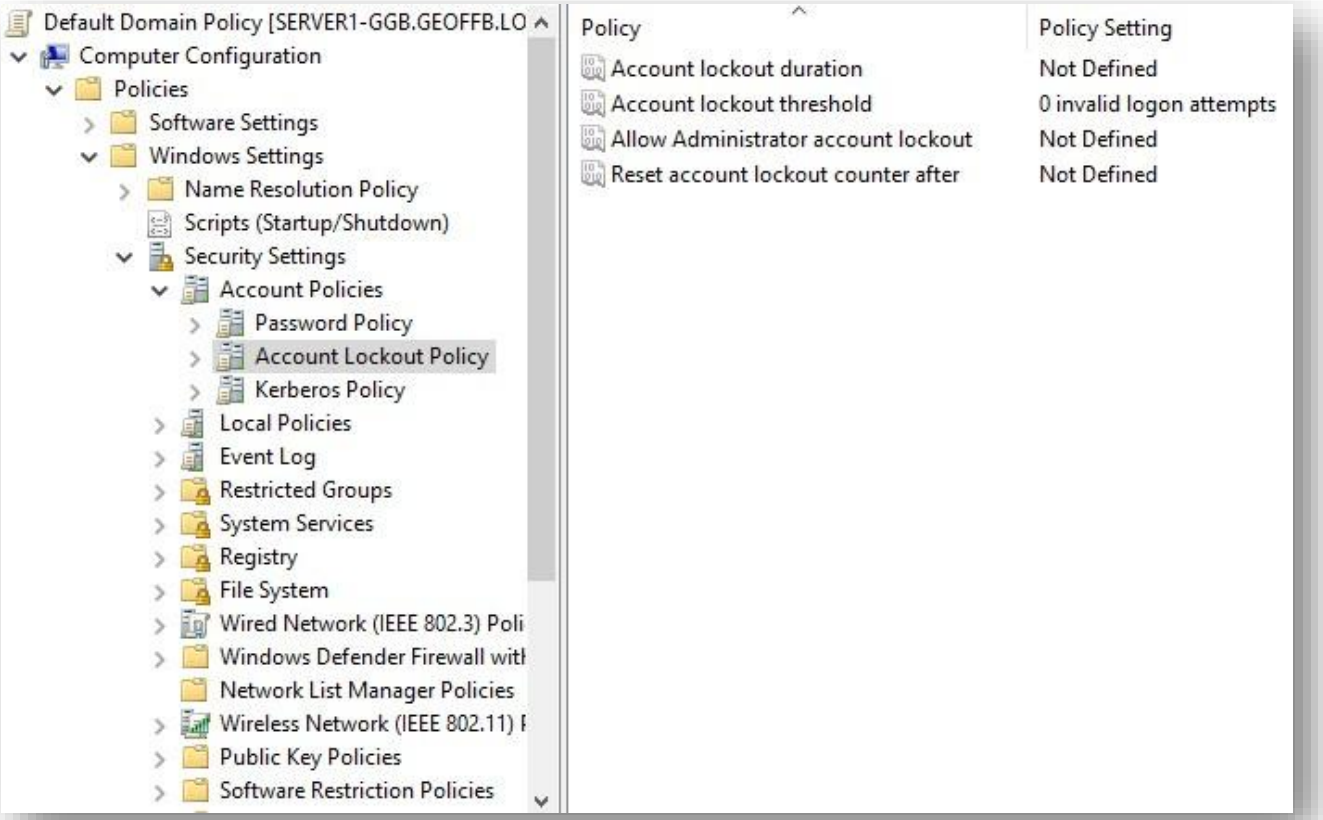
- Account lockout happens when, after several failed logon attempts by a single user, the system assumes that an attacker is attempting to compromise the account by discovering its password and locks out the account so no further logon attempts can be made.
- Domain account lockout policies determine how many invalid logon attempts can be made in a given period of time before the account becomes locked out.
- The policies also determine whether an administrator must be contacted to unlock the account, or if it becomes unlocked after a time period.
- These policy settings can only be applied on the Domain Container; applying them to sites or OUs will have no effect.

Account Lockout Policy Options



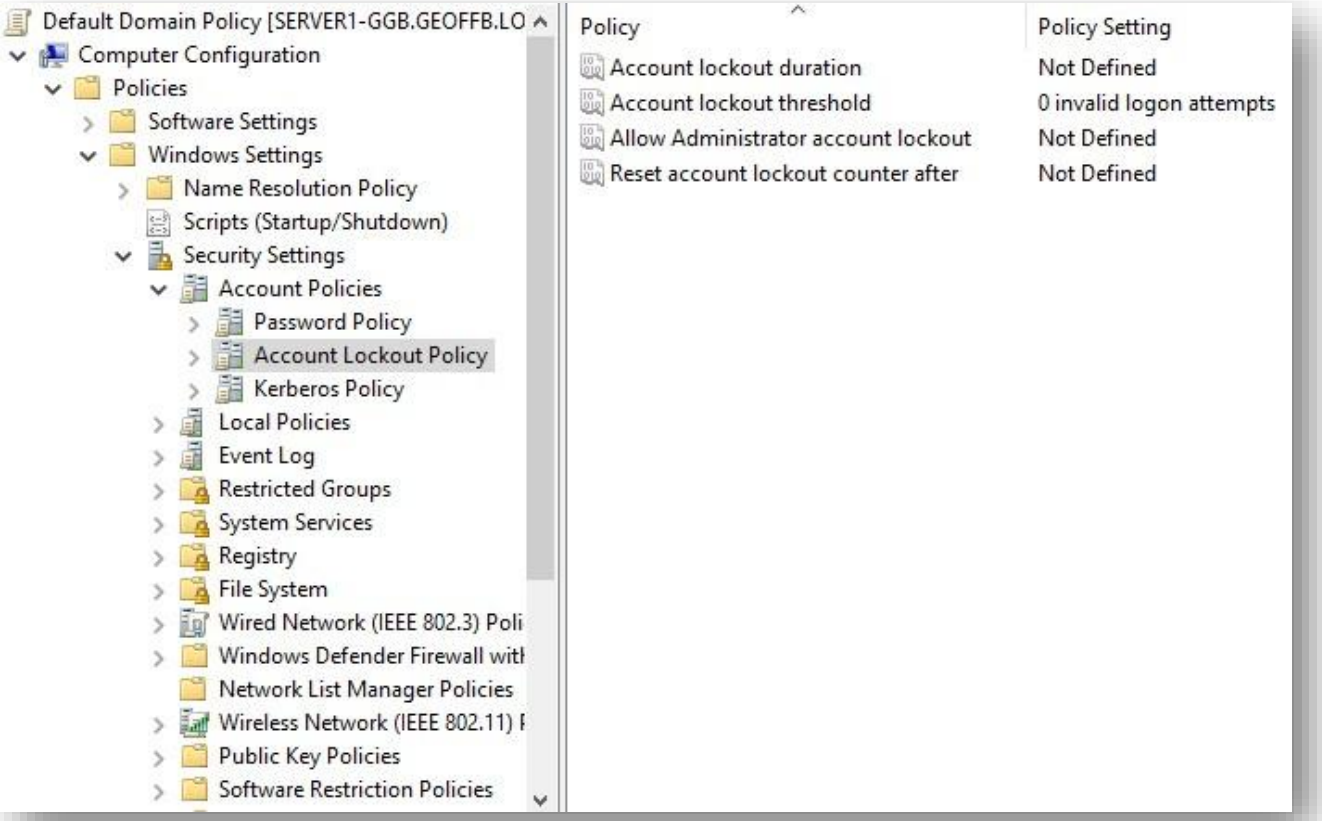
- Account Lockout Threshold
  - Enables you to set a limit to the number of unsuccessful tries to log onto an account
  - Values from 0 to 999 are valid (0 disables account lockout)
  - Low values can cause account lockout due to normal user error

Account Lockout Policy Options



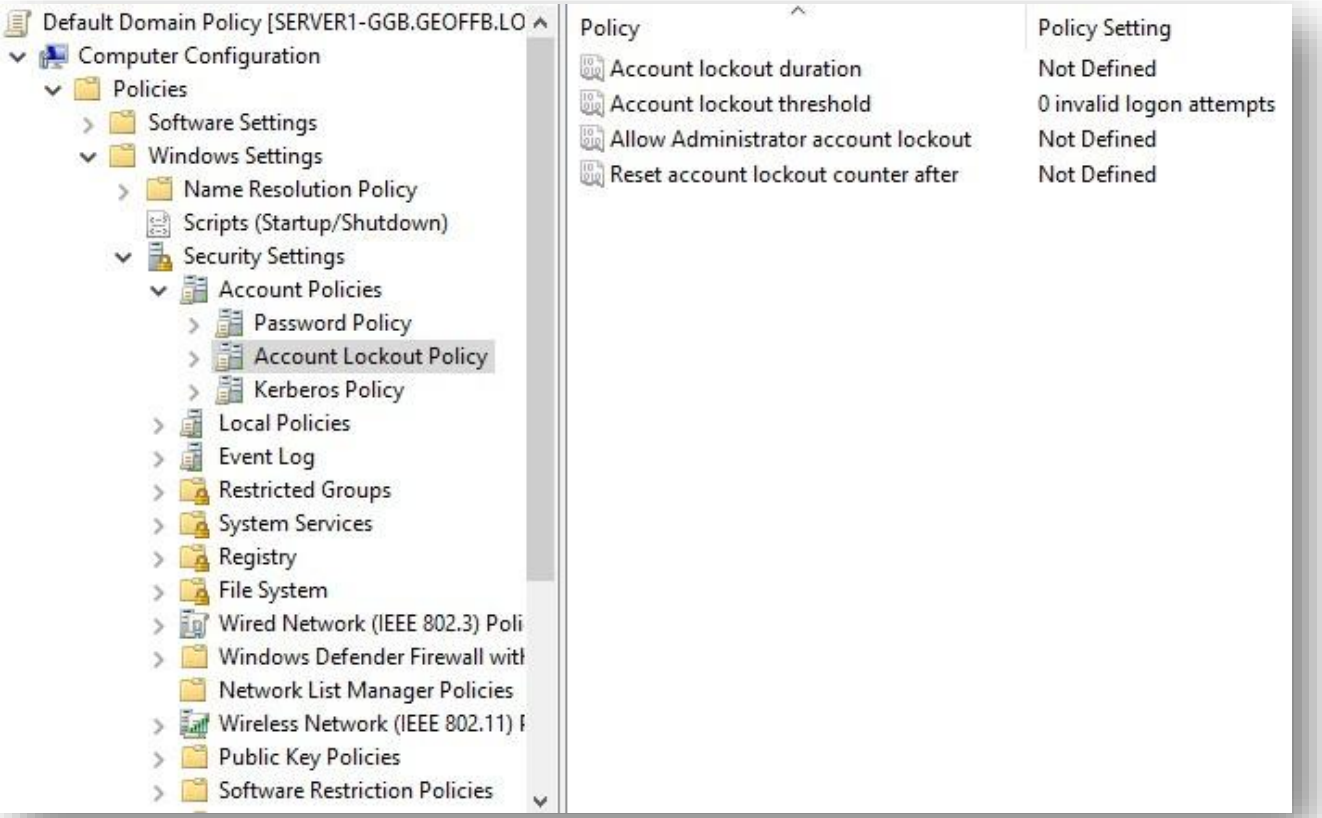
- Account Lockout Duration
  - Permits you to specify in minutes how long the system will keep an account locked out after reaching the specified number of unsuccessful log on attempts
  - Values from 0 to 99,999 minutes are valid (0 requires an administrator be contacted to unlock the account)
  - Low values provide moderate security protection, while minimizing user inconvenience

Account Lockout Policy Options



- Allow Administrator account
  - This security setting determines whether the built-in Administrator account is subject to account lockout policy.

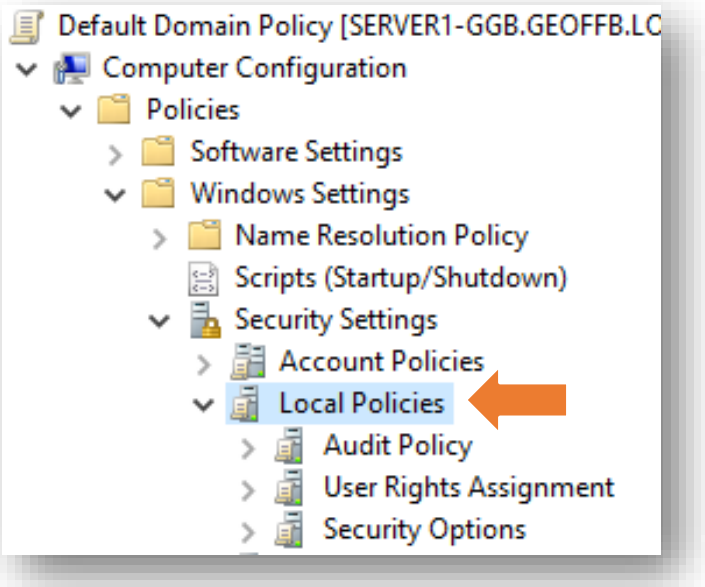
Account Lockout Policy Options



- Reset account lockout count after:
  - Period of time that must pass after invalid logon attempt before the lockout counter resets to 0
  - Values between 1 to 99,999 minutes and must be less than or equal to the value of the Account Lockout Duration

Local Policies

- Auditing, User Rights, and Securing the computing environment.
- Account policies focus on:
  - Audit Policy (Advanced Audit Policy)
  - User Rights Assignment
  - Security Options



Group Policy Processing order and Inheritance

Group Policy Processing order and Inheritance

- All policy settings, from all GPOs, are written to registry.
- Polices sources:
  - Local polices configured through GPEdit.msc.
  - Polices linked to sites
  - Polices linked to domain containers (affect all computers and users to which they apply).
  - Polices linked to OUs (affect all computers and users to which they apply).

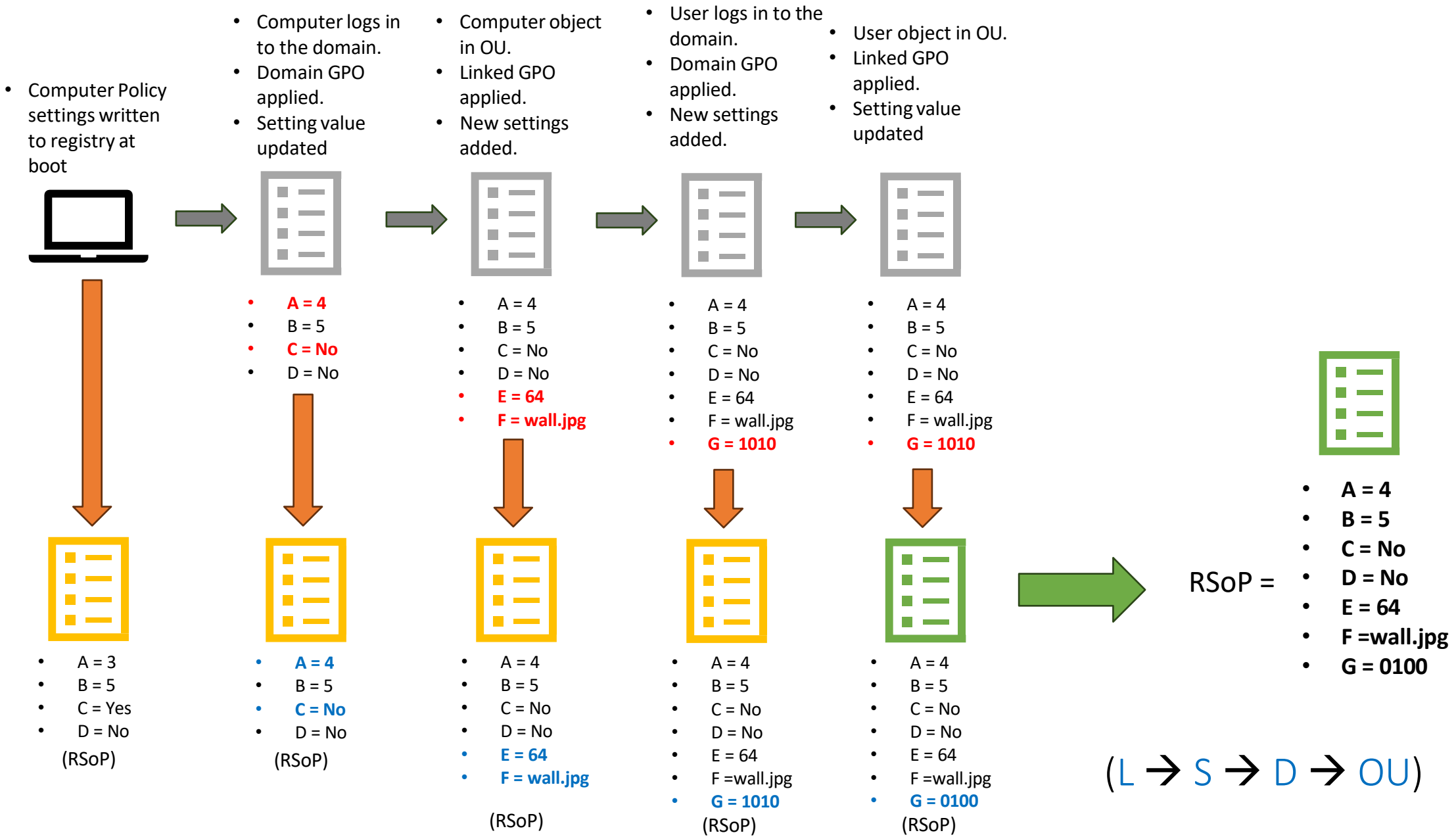
1. All GPO settings are written to registry once read.
2. Policy settings that are applied *first* will be **overwritten** by polices that are applied later.
3. Last GPO to be applied “wins”.

(we’ll talk about how conflicting settings are handled later)

(L → S → D → OU)

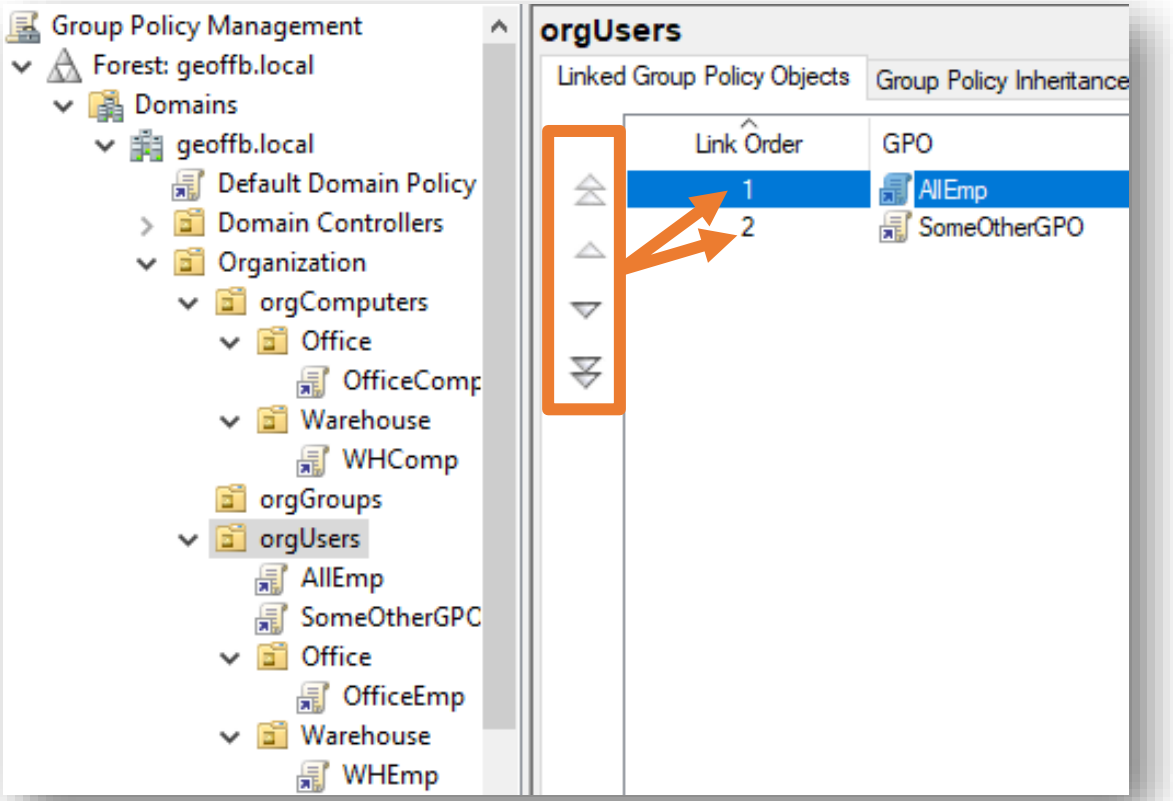
- Policies are applied in the following order:
  (Remember, the settings could be for computers or users. This process happens for both users and computers).
  1. Local (computer)
  2. Site
  3. Domain
  4. OU
- Group Policy settings will be cumulative if they do not conflict
- "Last to apply wins" can be interpreted as "the GPO setting closest to the leaf object wins".
- If a setting exists in both Computer and User settings, the Computer setting *usually* wins (though there are exceptions).

Resultant Set of Policy (RSOP)



Group Policy Inheritance

- If multiple GPOs are linked to a single site, domain, or OU, the administrator will have to choose which one has the higher priority by changing the order of GPOs by highlighting a GPO and clicking Up and Down buttons.
- GPOs with a *smaller precedence number* are processed **last** and take precedence over GPOs with higher numbers.



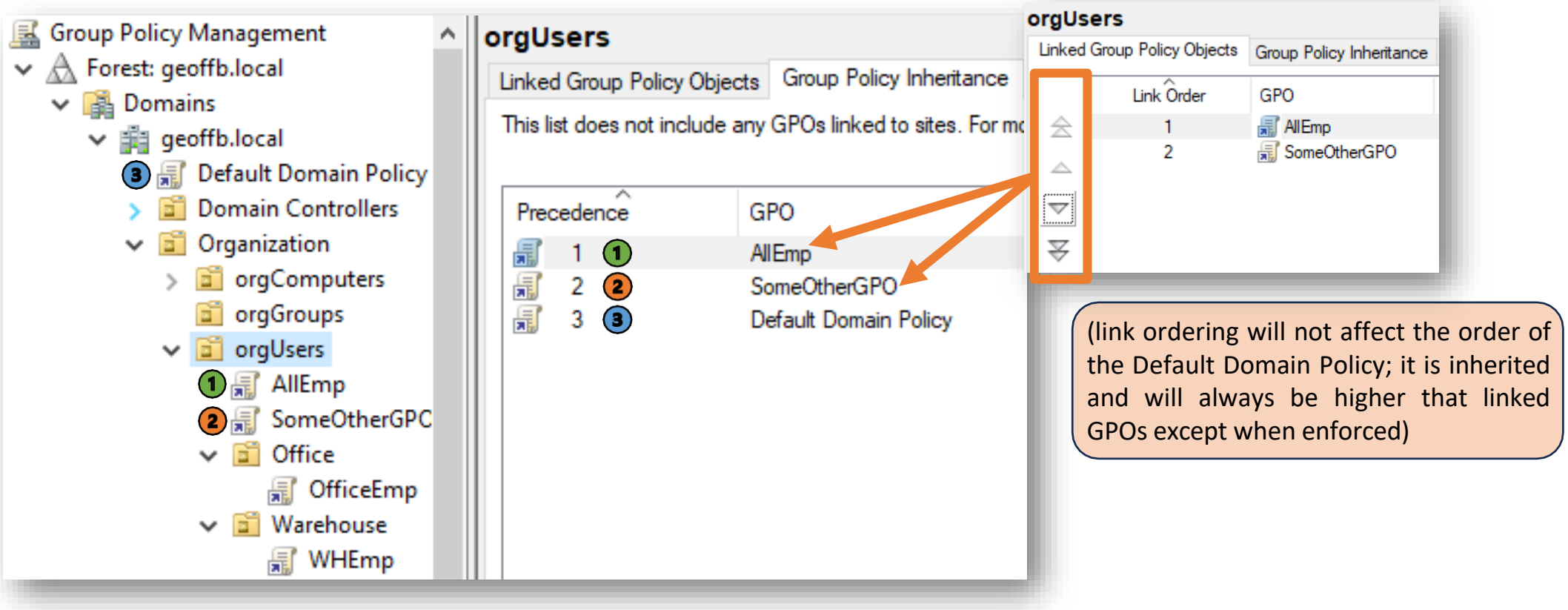
(someOtherGPO is processed first. AllEmp is processed last)

Do this…

- Link WHComp GPO to orgComputers/Warehouse.
- On the orgComputers/Warehouse set the WHComp GPO to a higher precedence number

Group Policy Inheritance

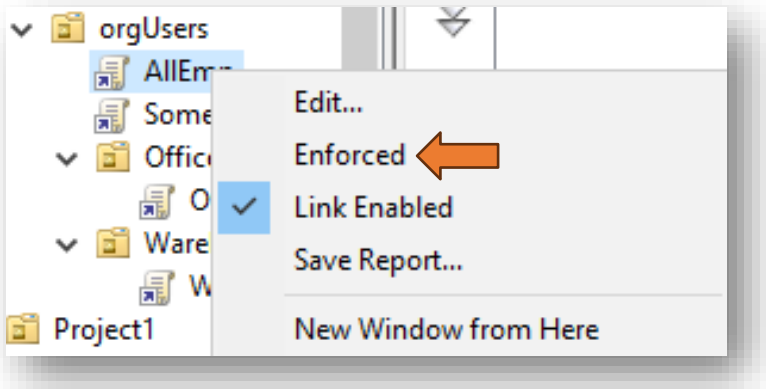
- Because of the parent/child relationship of Active Directory containers, we must also consider the policy objects that are being applied to the parent of the container.



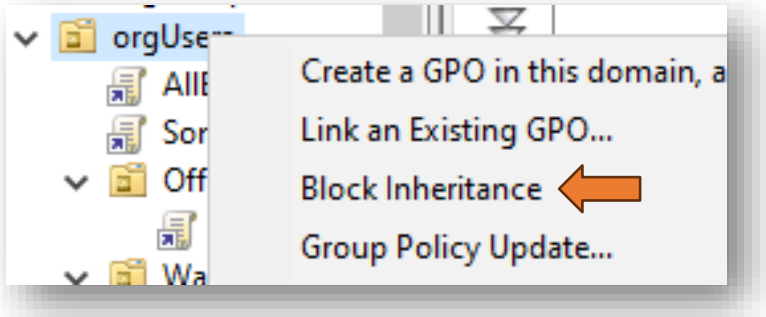
(link ordering will not affect the order of the Default Domain Policy; it is inherited and will always be higher that linked GPOs except when enforced)

Group Policy Inheritance

- Block Inheritance
  - Prevents policies from higher in the Active Directory tree being applied at lower levels of the tree.
- Enforce
  - Prevent policies at lower levels from overwriting policies applied at a higher level.
- When combining Enforce and Block Inheritance, Enforce takes precedence.
- If there is a conflict between the computer and user settings, the computer settings will take effect (in most cases).

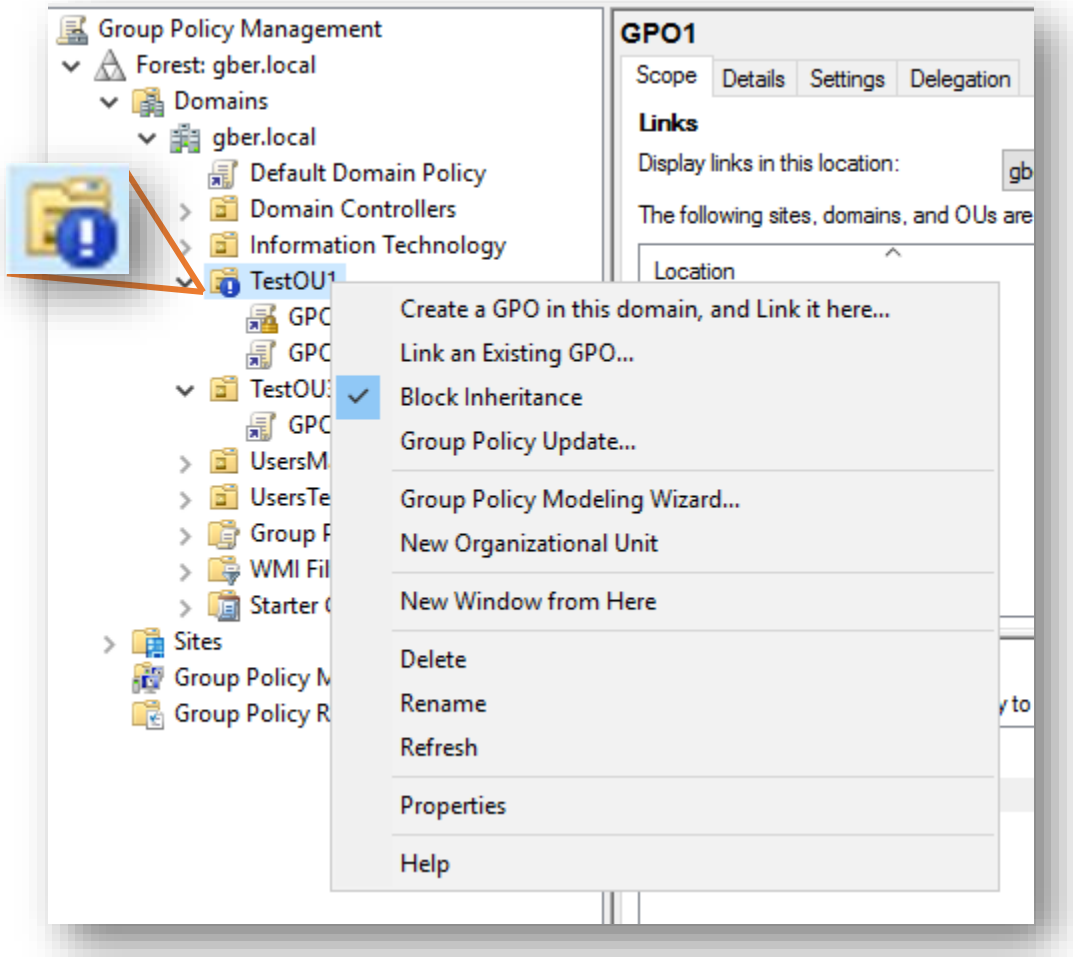


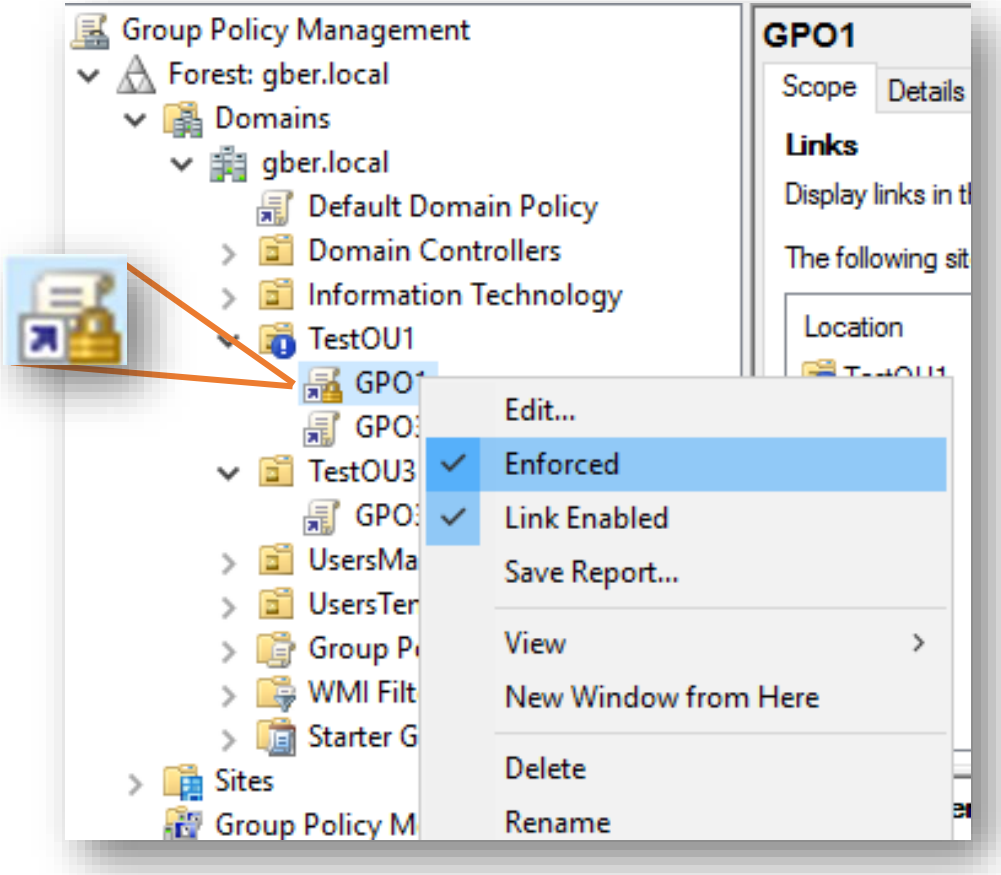
(Enforce option is on the GPO)



(Block Inheritance is on the OU)

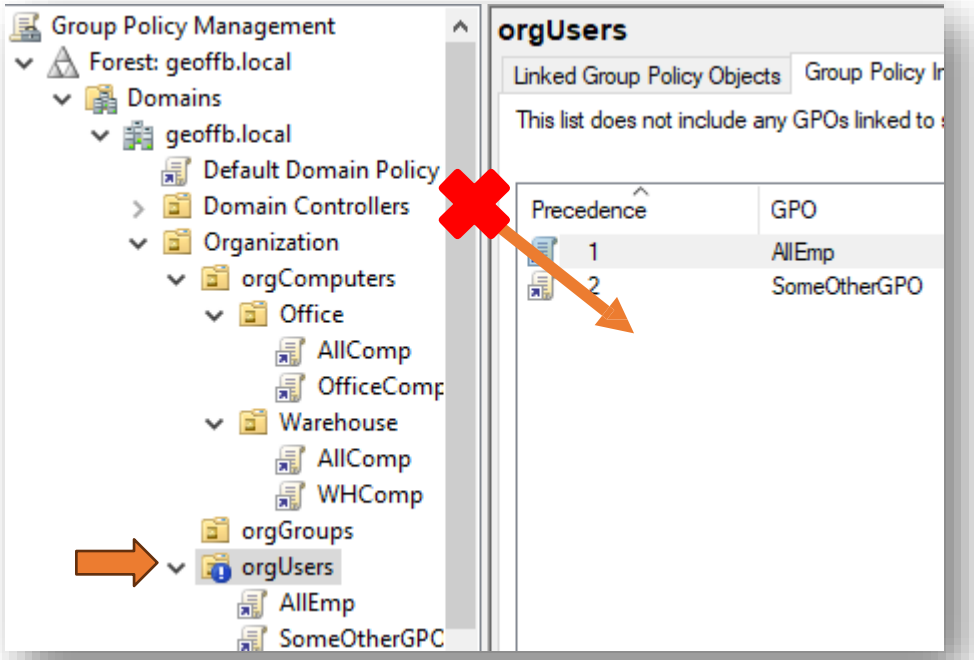
Block Inheritance is enabled on the OU, Enforced is enabled on the GPO (Start Tuesday)

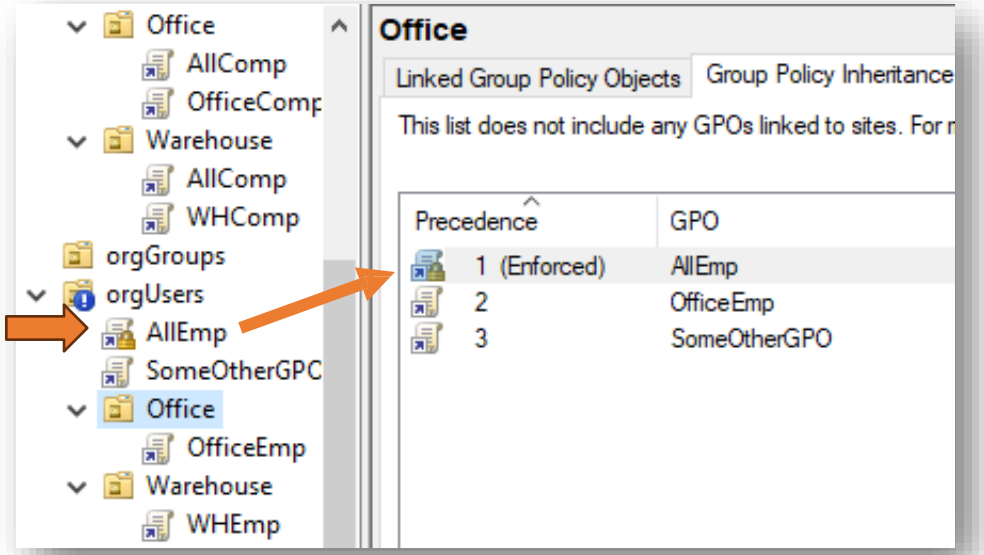




Do this…

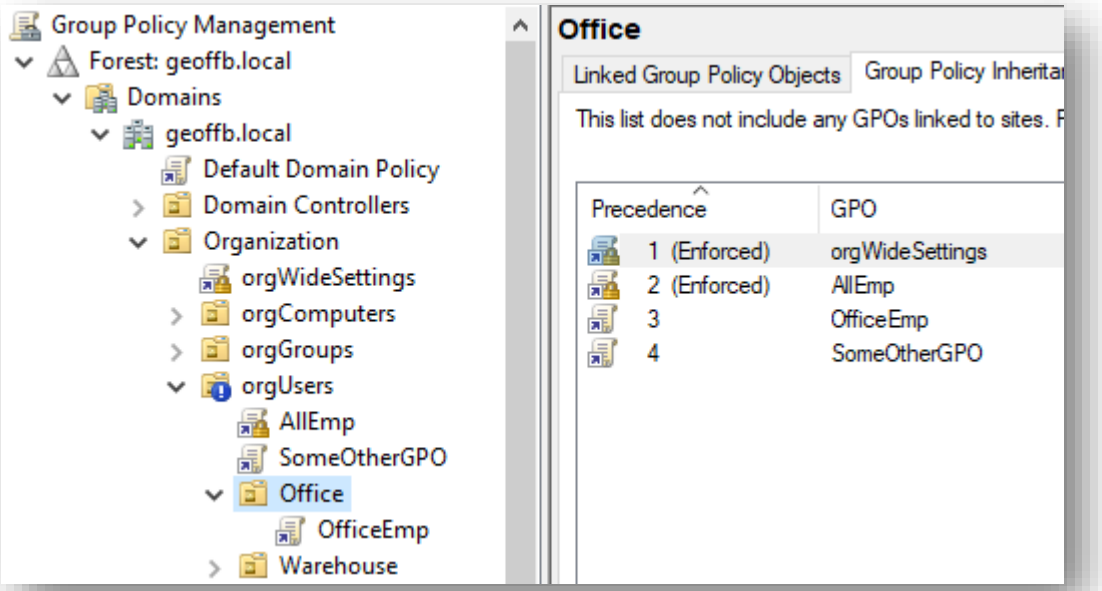
1. Stop the Default Domain Policy from applying to the orgUsers OU by blocking inheritance.
2. Link the OfficeEmp GPO orgUsers/Office.
3. Ensure the AllEmp has the highest precedence on the orgUsers/Office OU by enforcing AllEmp on orgUsers/Office.

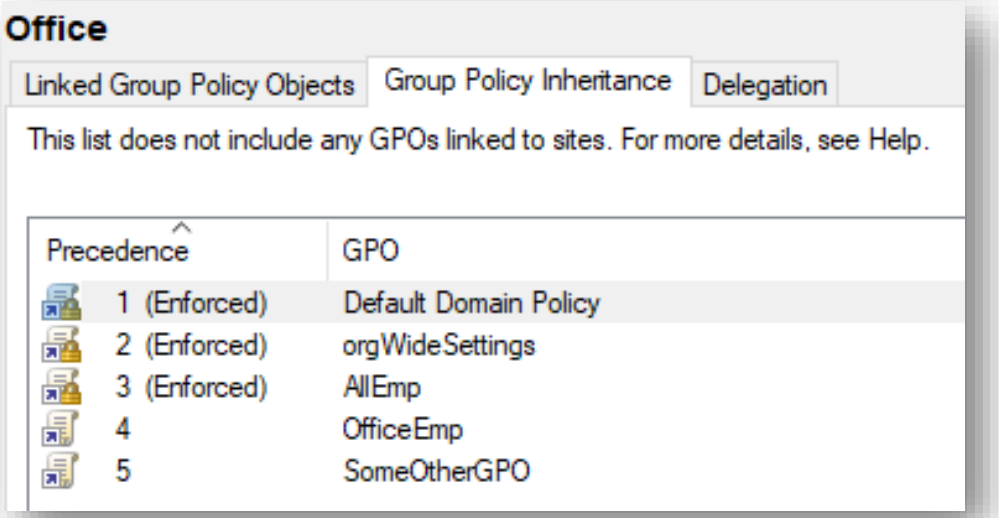




Do this…

4. Create and Link a GPO called orgWideSettings to Organization. Enforce the GPO.
5. Note the ordering of inherited objects on orgUsers/Office.
6. Enforce the Default Domain Policy.
7. Note the ordering of inherited objects on orgUsers/Office.
8. Unenforce the Default Domain Policy.



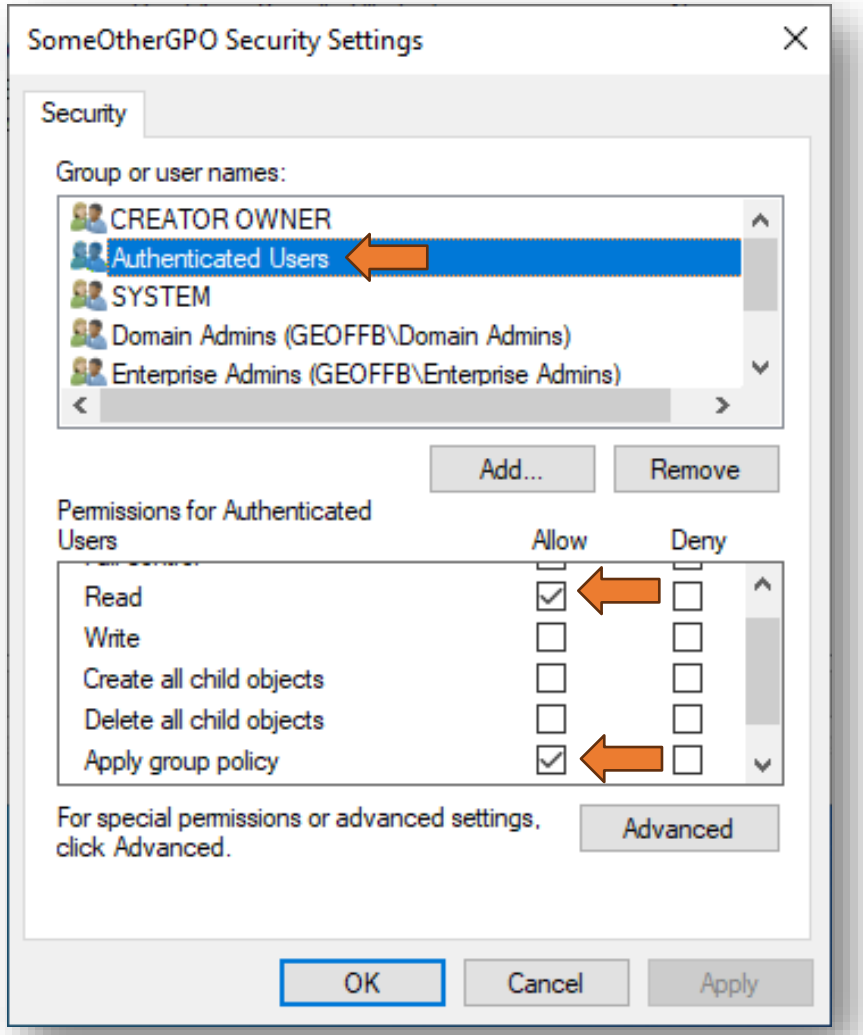


Rules for Determining Effective Policy (Review)

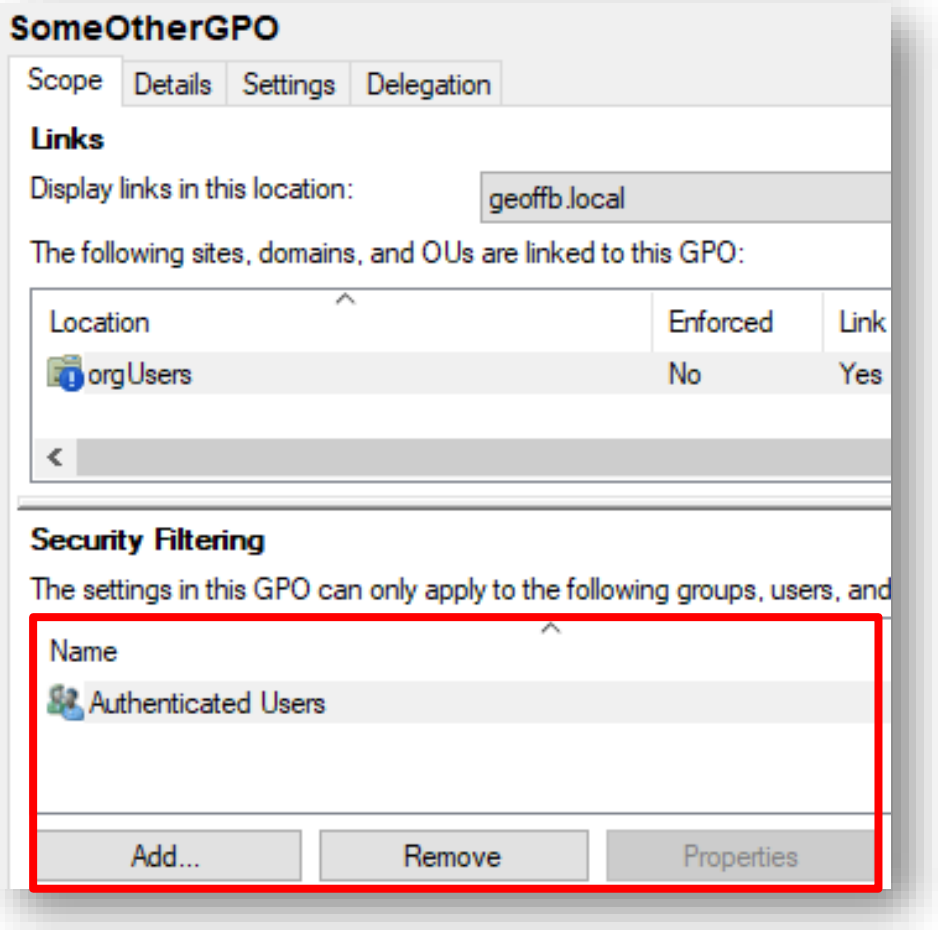
- Computer setting overrides user settings *in most cases* (if there is a conflict). If a policy is *shared* by both User and Computer settings, the policy will usually explain how a conflict is handled.
- LSDOU order of precedence.
- Site Policies apply to both computers and users (regardless of Domain or OU).
- Lowest Link order wins (lowest number) if multiple GPOs exist on an OU (controlled by admin; default is order they were applied).
- Policy inheritance can be turned off VIA Block Inheritance.
- Policy inheritance (and blocking) can be overruled by Enforce (higher over lower). Enforcing a policy reverses the precedence logic. GPOs applied at higher levels will win, instead.
- If both Block and Enforce are used, Enforce wins.

GPO Filtering

- Limiting who is affected by the settings within a GPO
- For a security policy to apply to a user/group, they must have read and apply group policy permissions to the GPO. If a user lacks one of these permissions, the policy will not apply to them.
- By default, all GPOs *can* apply to all users because Authenticated Users is granted Read and Apply Group Policy permissions.
- Remember: computers authenticate too. If this policy applies to computers, removing authenticated users will cause both users and computers to fail when attempting to apply the GPO.



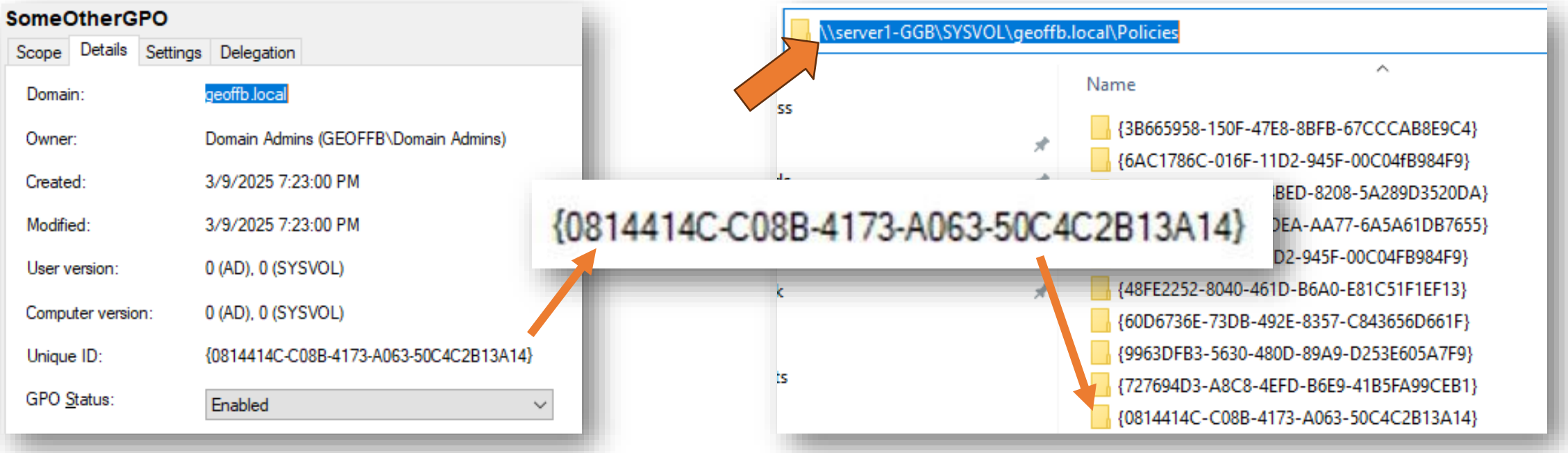
GPO Filtering



- You can assign read and apply group policy permissions in one of two ways:
  - Security Filtering section on the Scope tab of the GPO and add the user/group to the list.
- In Security Filtering (scope tab), any principal that appears in this list will be able to apply the policy.

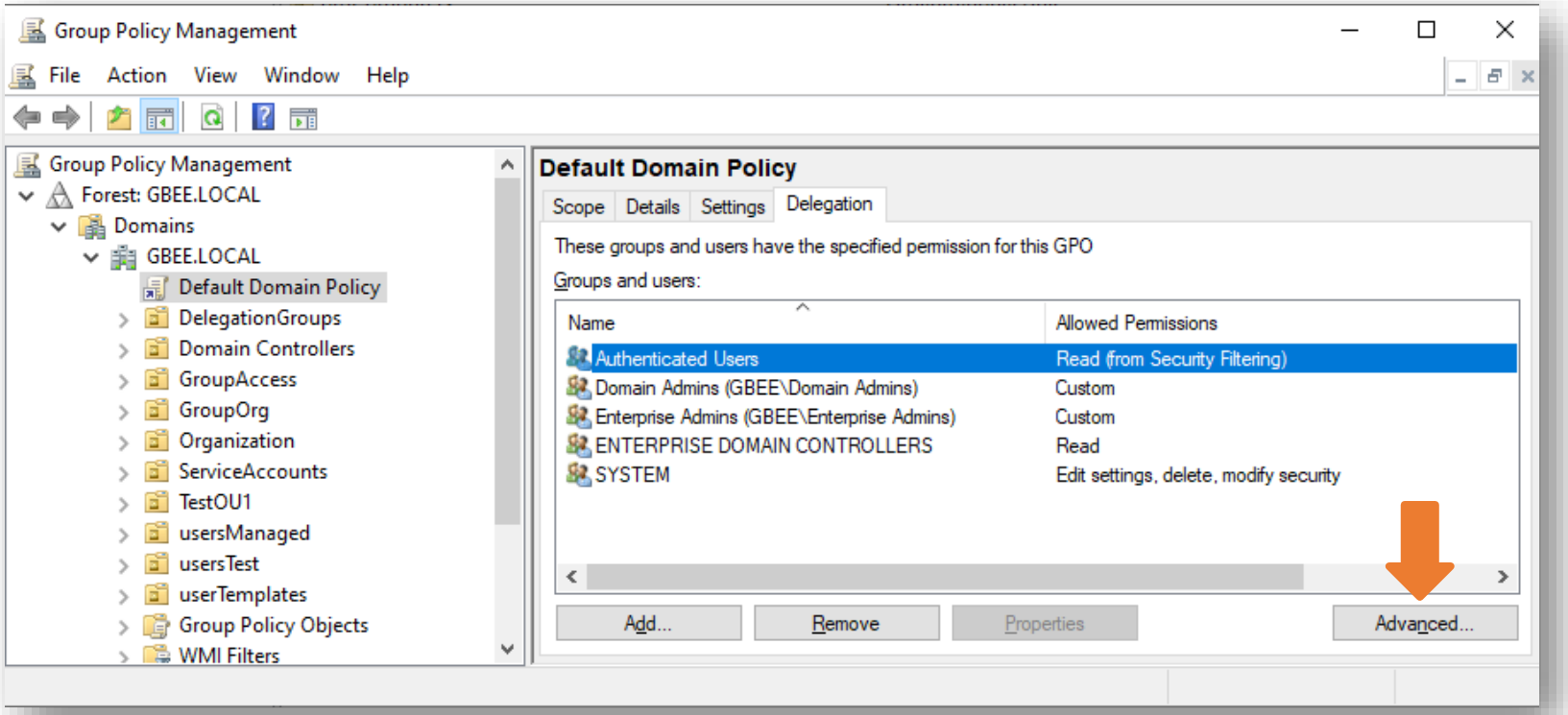
GPO Filtering

- When a user or computer logs into the domain, it must read its policies from the \\<domainController>\SYSVOL share.
- Having Read and Apply policy permissions are necessary for this to happen.

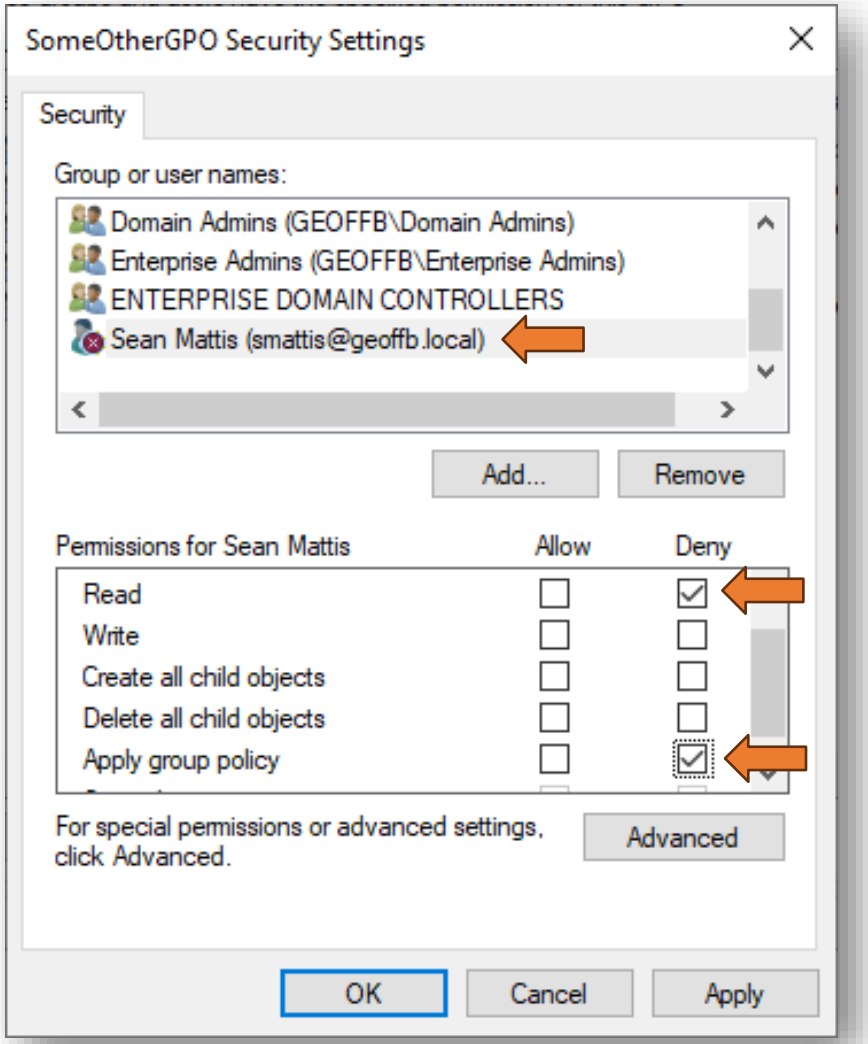


GPO Filtering

- Delegation tab of the GPO, Advanced button (bottom right corner), add the user/group and ensure read and apply group policy are selected.

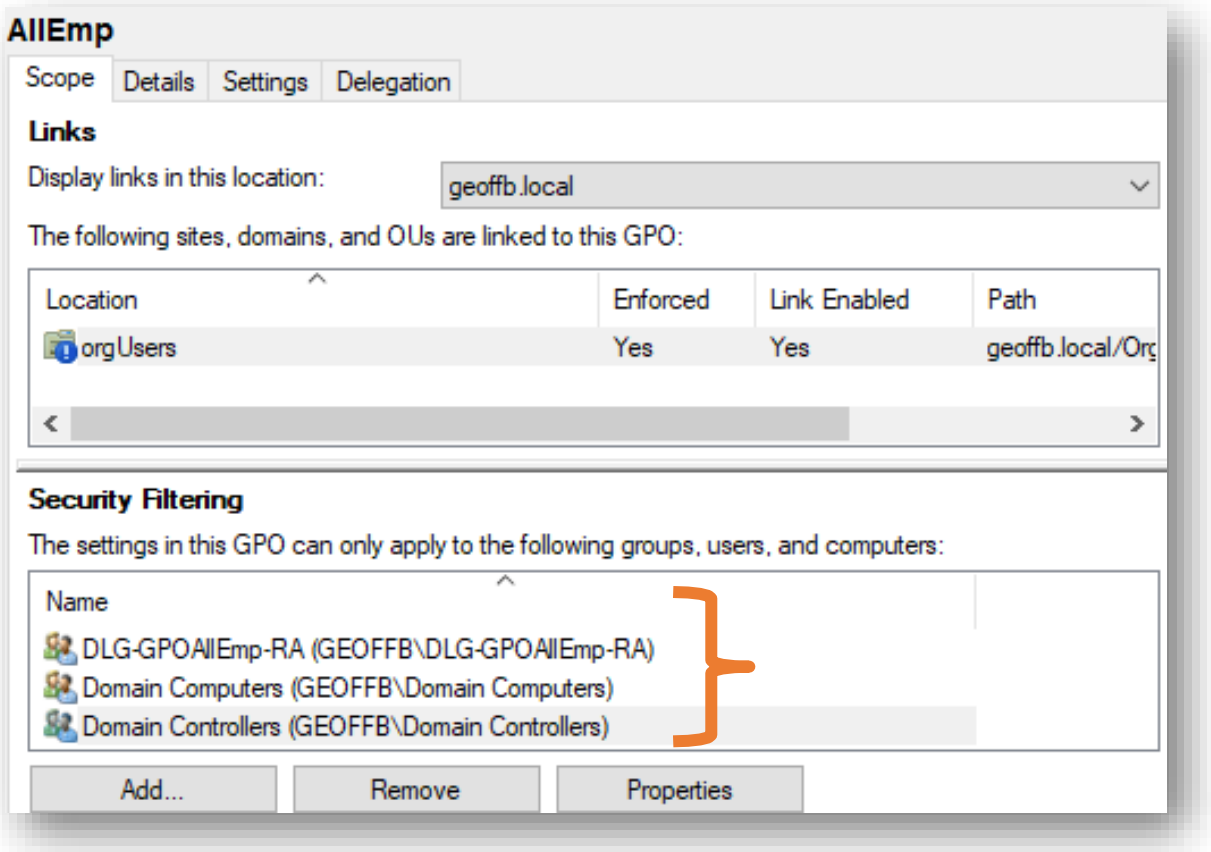


GPO Filtering (first way)



- To prevent restrictive policies from applying to administrators or other accounts for which they are not intended:
  - Add the principal account or group to the ACL using the deletegaion tab → Advanced Buttong
  - Configured the Read/Apply Group Policy ACE to Deny for the specific user/group that you want to exclude from the group policy

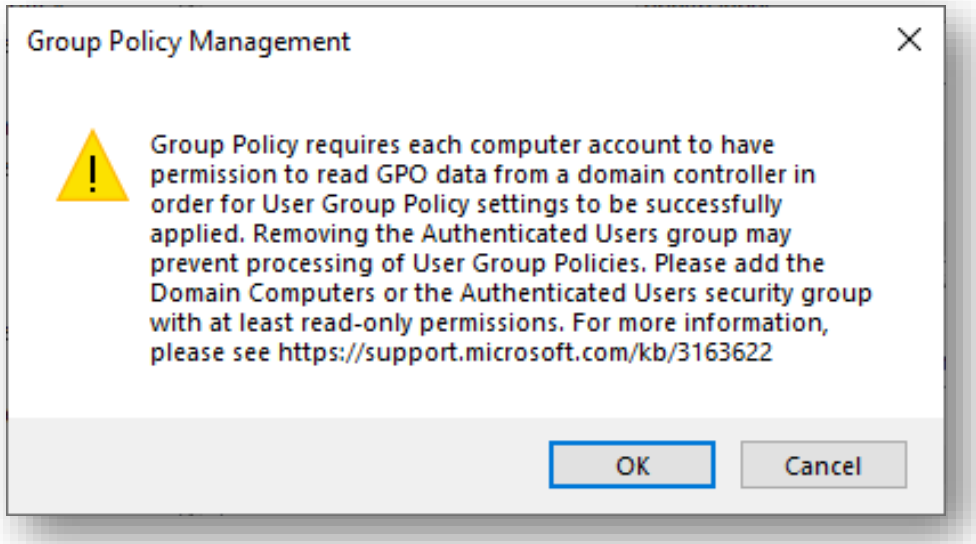
GPO Filtering (second way)

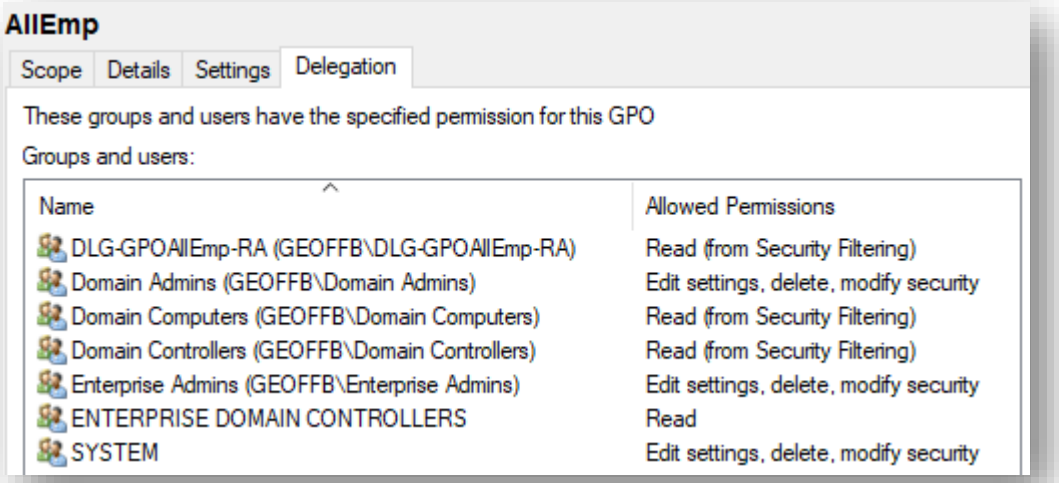


- To prevent restrictive policies from applying to administrators or other accounts for which they are not intended:
  - Remove authenticated users from Security Filtering
  - Add the principal you want to give Read/Apply to.
  - Add either domain computers or domain controllers security groups to security filtering (so the computers can still read the policies).

Do this…

1. In the orgGroups/userGroups create a global group called GG-AllEmp
2. In the orgGroups/accessGroups create a domain local group DLG-GPOAllEmp-RA
3. Add all orgUsers to the GG-AllEmp group
4. Add the GG-AllEmp to the DLG-GPOAllEmp-RA group.
5. Remove Auth Users from Security filtering.
6. Add DLG-GPOAllEmp-RA, Domain Computers, and the Domain Controllers security groups to security filtering.
7. Look at the Delegation Tab.
8. Attempt to remove Auth Users from the Scope Tab under Security filtering. Note the message.





Auditing (Start Day 3)

Computer Configuration\Windows Settings\Security Settings\Advanced Audit Policy

- Tracking the success or failure of events associated with an object
- To configure auditing, start by configuring a group policy for auditing.
- The following audit policies are located in the Computer Configuration\Windows Settings\Security Settings\Advanced Audit Policy
- Legacy Audit Policy Settings are found at Computer Configuration\Policies\Windows Settings\Security Settings\Local Polices\Audit Policy

```
auditpol /get /category:* | more
```

auditpol /get /category:* | more (default)

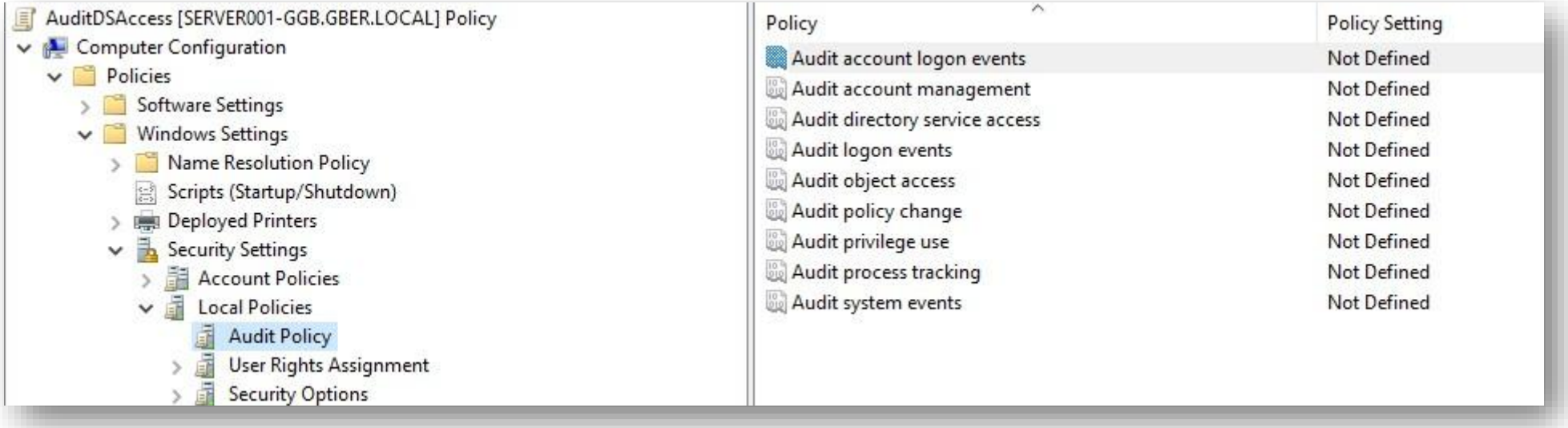
```
PS C:\> auditpol /get /category:* | more
System audit policy
Category/Subcategory                        Setting
System
  Security System Extension                 No Auditing
  System Integrity                          Success and Failure
  IPsec Driver                              No Auditing
  Other System Events                       Success and Failure
  Security State Change                     Success
Logon/Logoff
  Logon                                     Success and Failure
  Logoff                                    Success
  Account Lockout                           Success
  IPsec Main Mode                           No Auditing
  IPsec Quick Mode                          No Auditing
  IPsec Extended Mode                       No Auditing
  Special Logon                             Success
  Other Logon/Logoff Events                 No Auditing
  Network Policy Server                     Success and Failure
  User / Device Claims                      No Auditing
  Group Membership                          No Auditing
Object Access
  File System                               No Auditing
  Registry                                  No Auditing
  Kernel Object                             No Auditing
  SAM                                       No Auditing
  Certification Services                    No Auditing
  Application Generated                     No Auditing
  Handle Manipulation                       No Auditing
  File Share                                No Auditing
  Filtering Platform Packet Drop            No Auditing
  Filtering Platform Connection             No Auditing
  Other Object Access Events                No Auditing
  Detailed File Share                       No Auditing
  Removable Storage                         No Auditing
  Central Policy Staging                    No Auditing
```

```
Privilege Use
  Non Sensitive Privilege Use               No Auditing
  Other Privilege Use Events                No Auditing
  Sensitive Privilege Use                   No Auditing
Detailed Tracking
  Process Creation                          No Auditing
  Process Termination                       No Auditing
  DPAPI Activity                            No Auditing
  RPC Events                                No Auditing
  Plug and Play Events                      No Auditing
  Token Right Adjusted Events               No Auditing
Policy Change
  Audit Policy Change                       Success
  Authentication Policy Change              Success
  Authorization Policy Change               No Auditing
  MPSSVC Rule-Level Policy Change           No Auditing
  Filtering Platform Policy Change          No Auditing
  Other Policy Change Events                No Auditing
Account Management
```

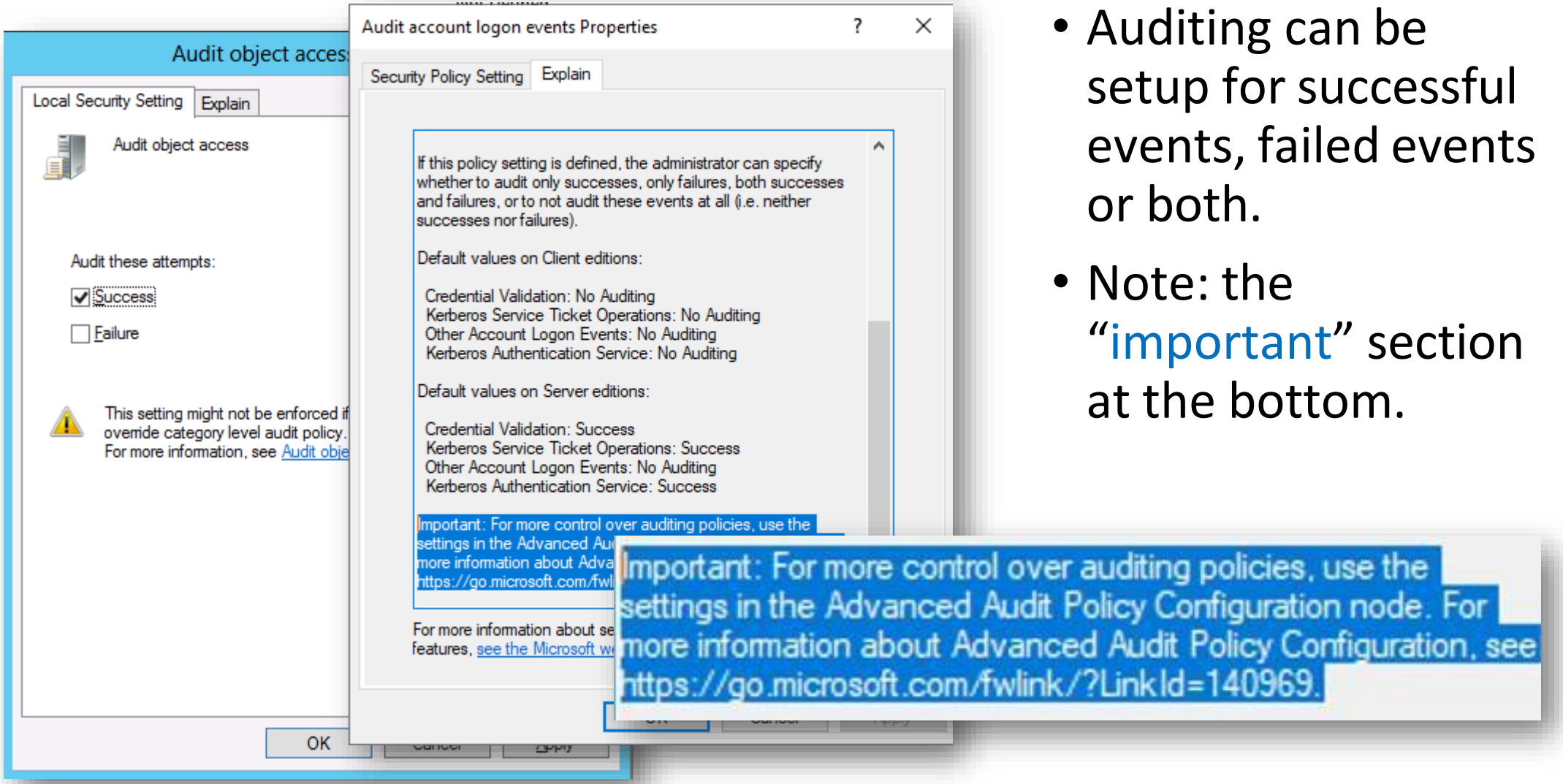
Security Policies

Computer Configuration\Policies\Windows Settings\Security Settings\Local Polices\Audit Policy

- Local Policies\Audit Policies
- Note: these policy settings are more general than advanced audit settings. Once Advanced Audit Policy is configured, these settings will be cleared (will not apply).



Auditing

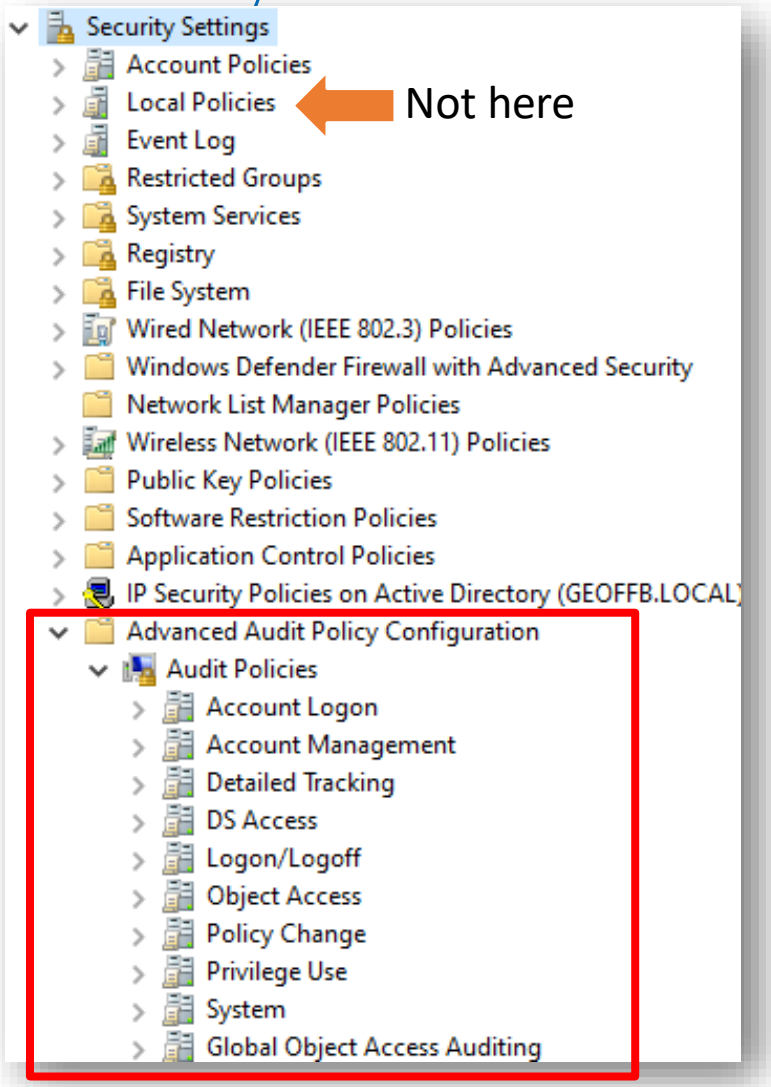


- Auditing can be setup for successful events, failed events or both.
- Note: the "important" section at the bottom.

Security Policies (Auditing)

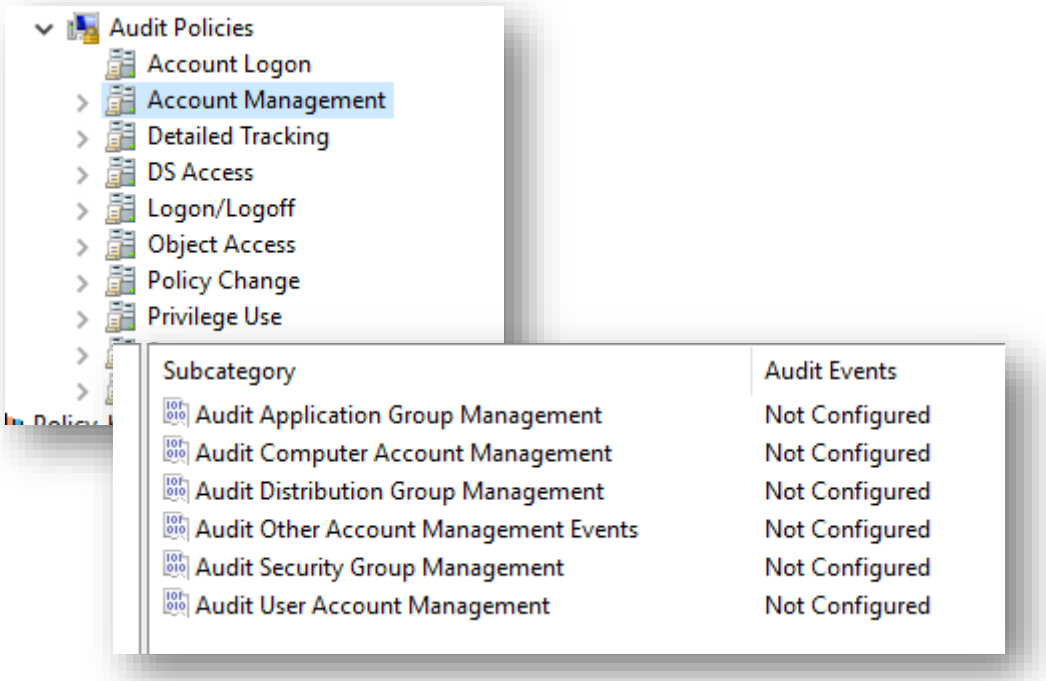
Computer Configuration\Windows Settings\Security Settings\Advanced Audit Policy

- Advanced Audit Policies provide more granular control over auditing.
- As a best practice, do not use Basic Auditing if you are going to also configure Advanced Auditing.
- Advanced auditing will clear basic auditing settings (if configured) but can have unexpected results if both are configured at the same time.
- For more in-depth auditing information please refer to the Microsoft Docs repository: https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/advanced-security-auditing-faq

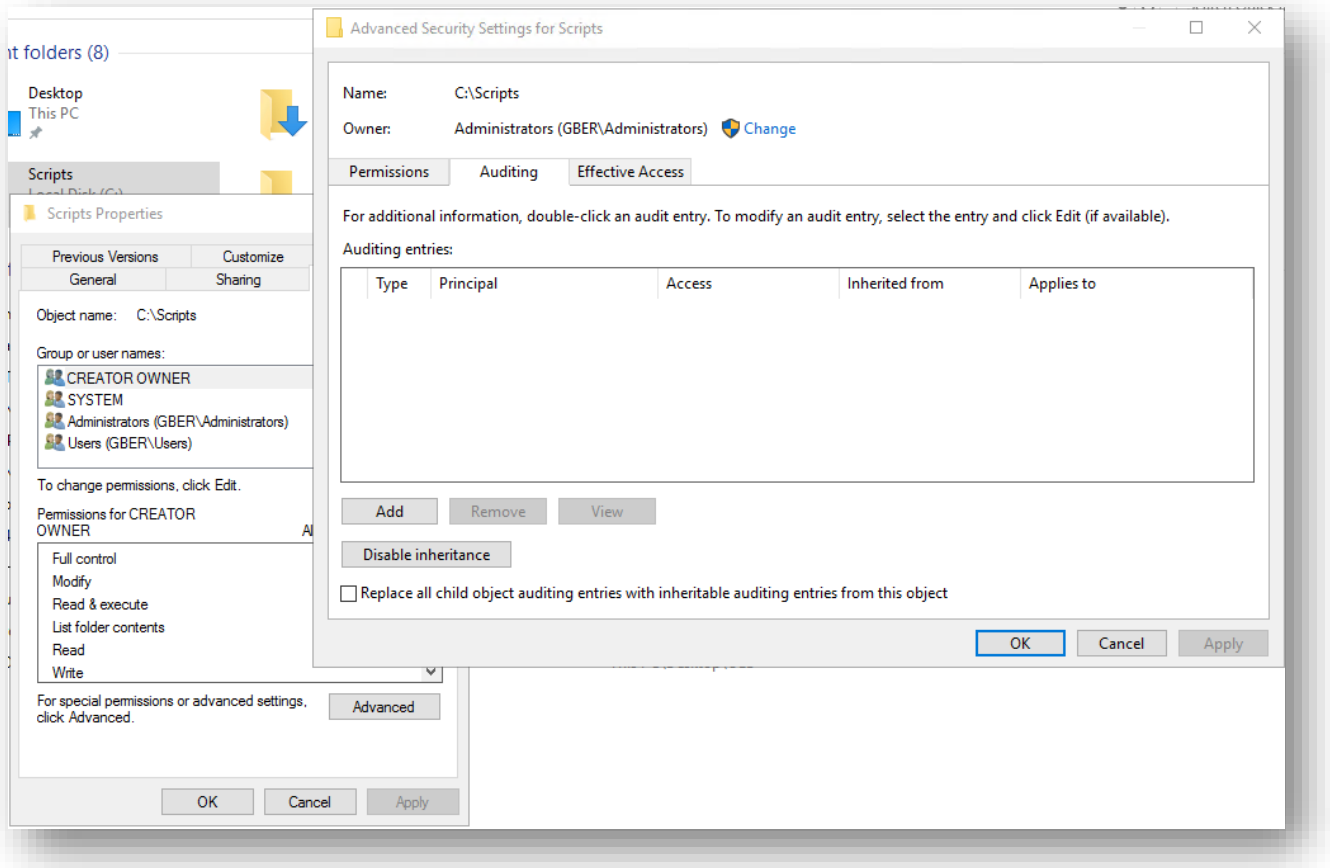


Advanced Auditing

- With advanced auditing we have access to 10 auditable categories.
  - Each category has subcategories with configurable settings.
- These settings specify the objects or events that will be audited.
- If auditing has been turned on for an object, we then must also configure the object itself (using the audit tab of the objects ACL).

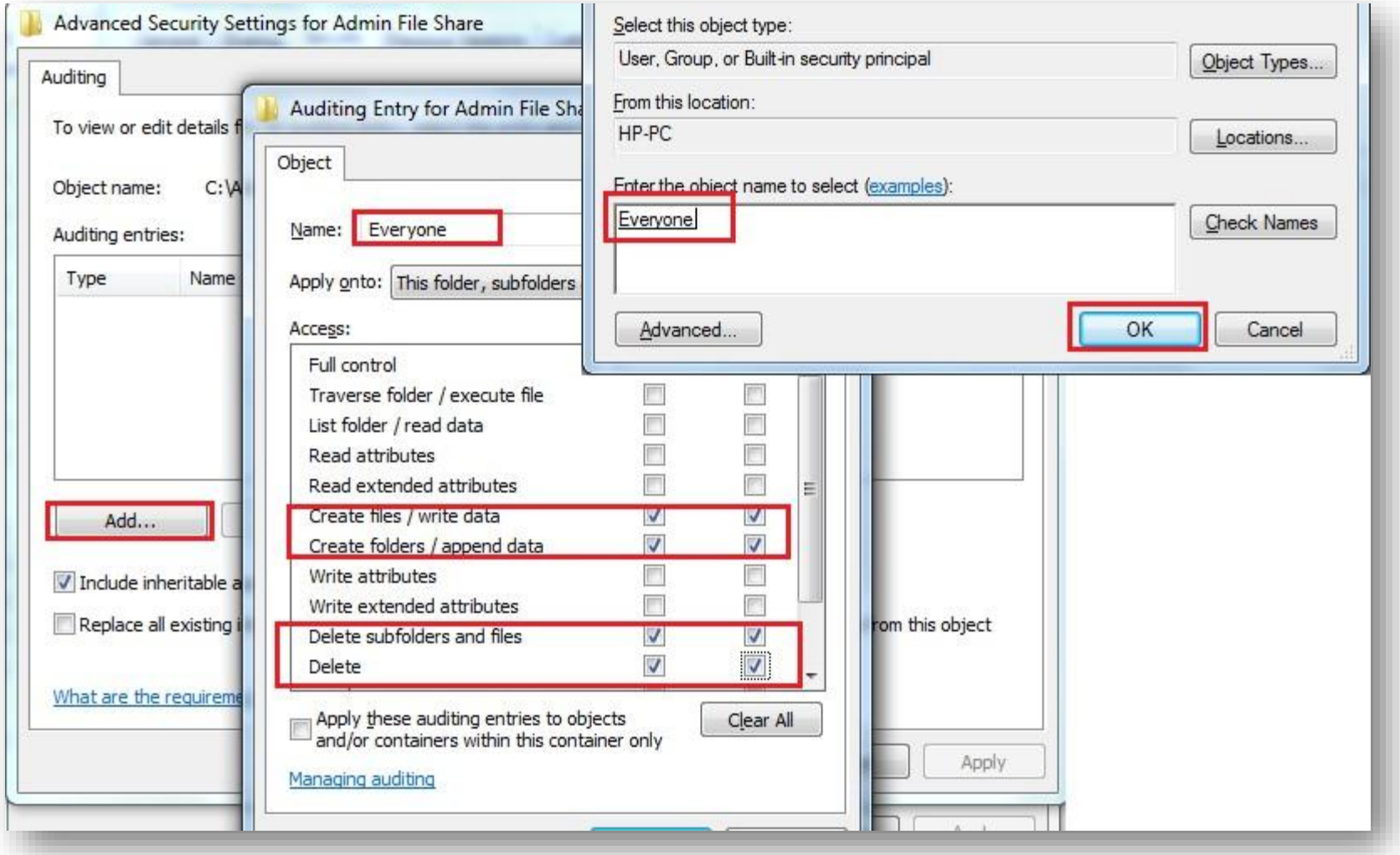


Auditing

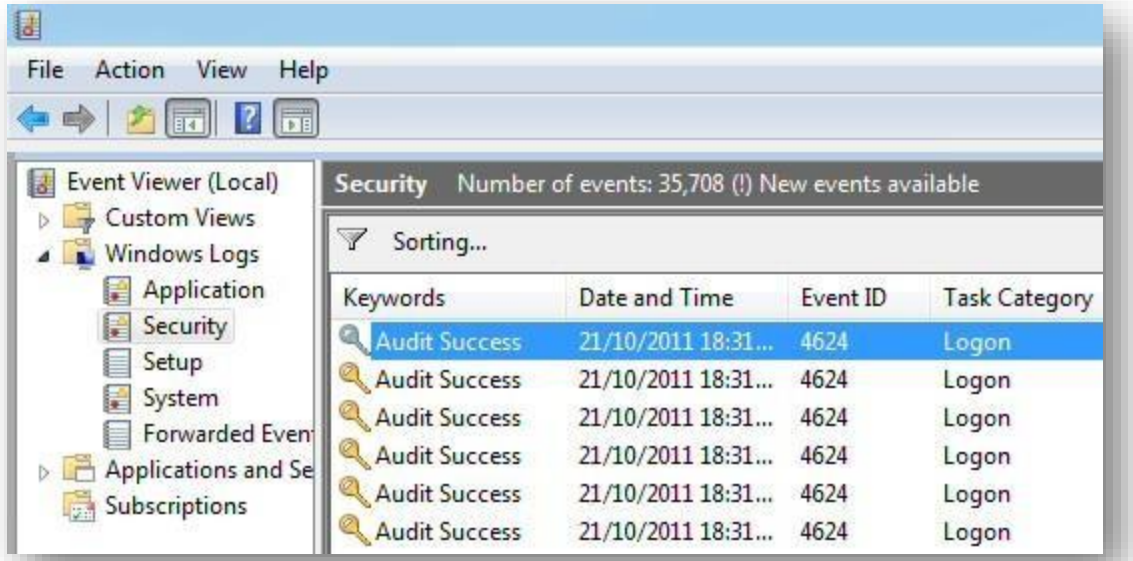


- If you configure auditing for object access or directory service access, you must then configure the individual files, folder and objects for which you want to audit.
- Set up auditing by accessing the Security tab of the object, click the Advanced button, select the auditing tab and configure the user/group to audit and what to audit about them.

Auditing

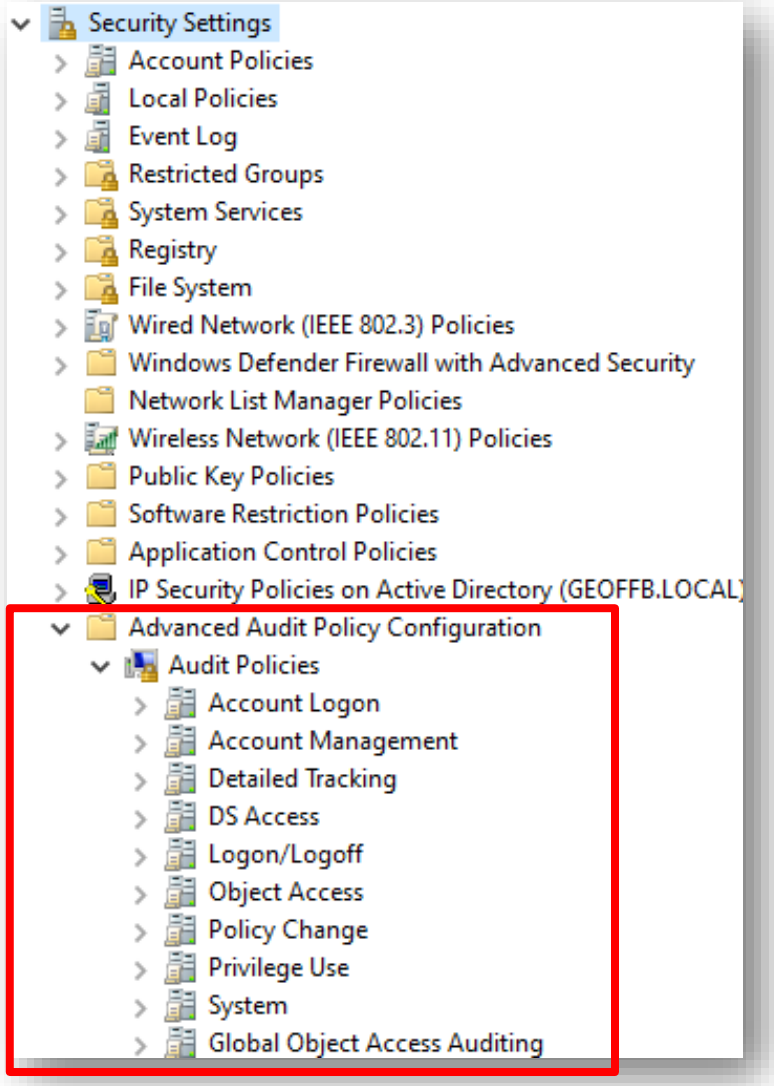

Advanced Security Settings for Admin File Share
Auditing
To view or edit details f
Object name: C:\A
Auditing entries:
Type
Name
Add...
Include inheritable a
Replace all existing i
What are the requireme
Auditing Entry for Admin File Sha
Object
Name: Everyone
Apply onto: This folder, subfolders
Access:
Full control
Traverse folder / execute file
List folder / read data
Read attributes
Read extended attributes
Create files / write data
Create folders / append data
Write attributes
Write extended attributes
Delete subfolders and files
Delete
Apply these auditing entries to objects and/or containers within this container only
Clear All
Managing auditing
Select this object type:
User, Group, or Built-in security principal
Object Types...
From this location:
HP-PC
Locations...
Enter the object name to select (examples):
Everyone
Check Names
Advanced...
OK
Cancel
rom this object
Apply

Auditing

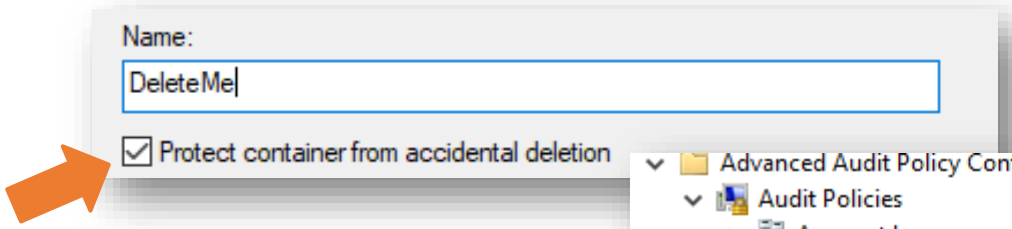


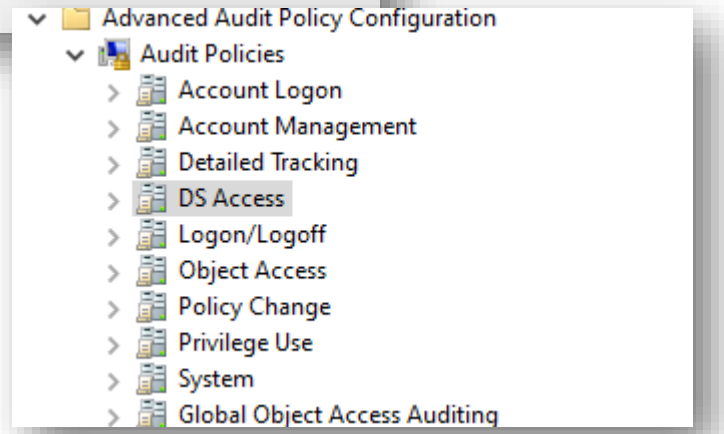
- Audited events are recorded in the Security log of Event Viewer.

- Account Logon
  - Audit the use of logon credentials.
- Account Management
  - Audit Account property changes and group membership changes for user and computer accounts.
- Detailed Tracking
  - Audit application and subsystem processes.
- DS Access
  - Audit Directory Service (AD object) access.
- Logon/Logoff
  - Audit the logon/off process, and the events surrounding logon (like enumerating polices and security group membership)
- Object Access
  - Audit the access of non-AD objects (like registry, SAM, local disks, certificates, etc).
- Policy Use Change
  - Audit the changes to policy including the audit policy itself.
- Privilege Use
  - Audit the privilege elevation and invocation.
- System
  - Audit the system for system events (start up/shutdown, specific events)
- Global Object Access Auditing
  - Comprehensive auditing of the File System or Registry.

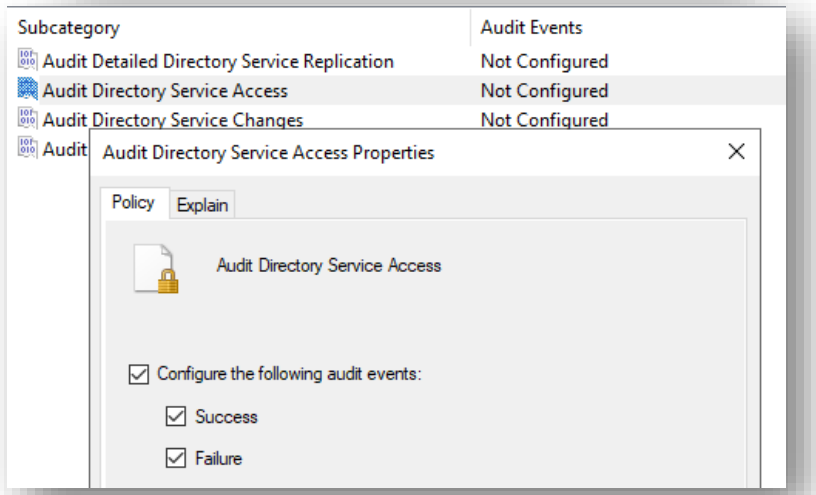


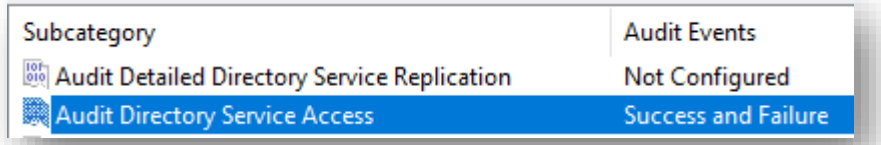
Do this…





1. Open ADUC. Create an OU called DeleteMe. Make sure you leave Protect container from accidental delete ON.
2. In GPMC, Create a Policy called AuditDSAccess, and turn on DS Access\Audit Directory Service Access for successes and failures.
3. Link the AuditDSAccess Policy to the Domain.
   `GPUDATE /FORCE`
4. Run `auditpol /get /category:* | more` to confirm settings.
5. In ADUC, turn on advanced view.
6. (on deleteme) Properties → Security tab → Advanced → Audit
7. Add → Principal (Everyone) → Type = All → Defaults + Delete and Delete Subtree
8. Open Event Viewer and Clear the Security logs.
9. Try to delete DeleteMe then check eventvwr for the 4662 Delete log

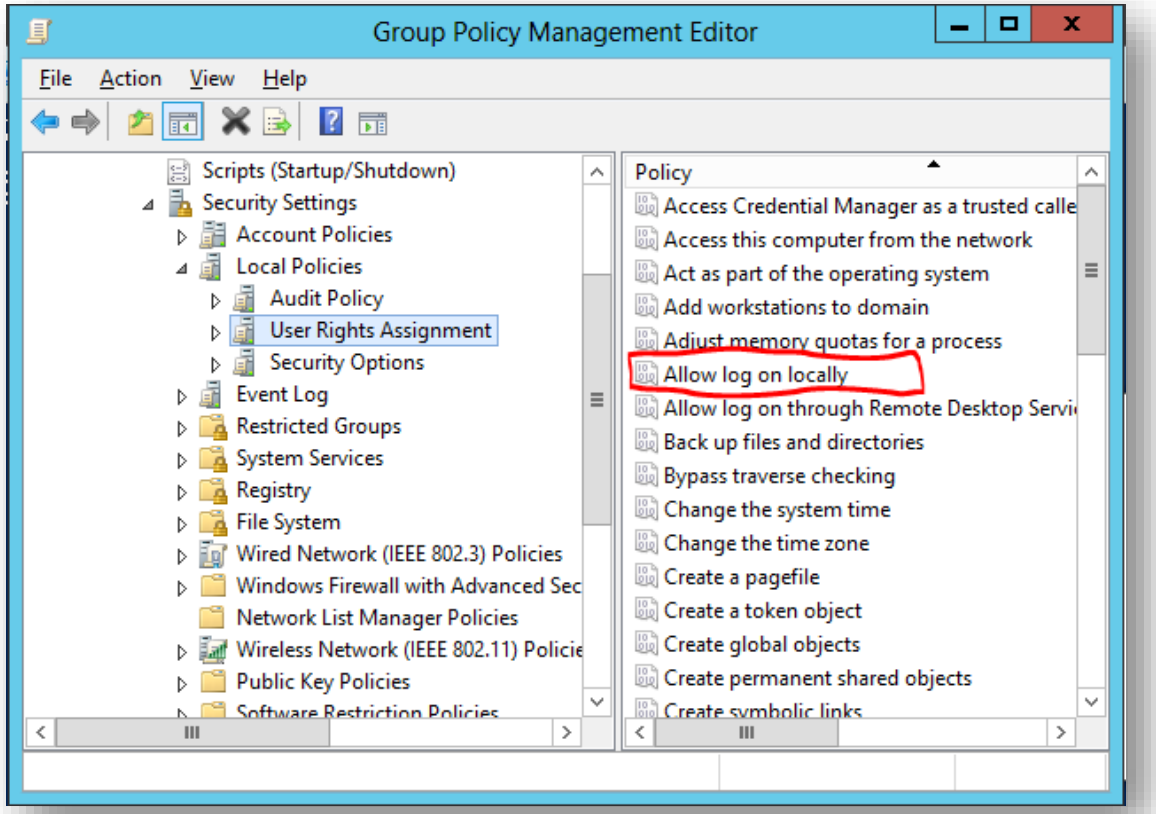




Security Policies

Computer Configuration\Policies\Windows Settings\Security Settings\Local Polices\User Rights Assignment

- Grant users/group's ability to perform system-wide tasks, such as shut down the system or logon locally.



Key User Rights Assignment Settings

- Allow Logon locally
  - Determines which users can log on to the computer.
- Deny log on locally
  - This security setting determines which users are prevented from logging on at the computer
  - This policy setting supersedes the Allow log on locally policy setting if an account is subject to both policies.
- Change the system time
  - This user right determines which users and groups can change the time and date on the internal clock of the computer.

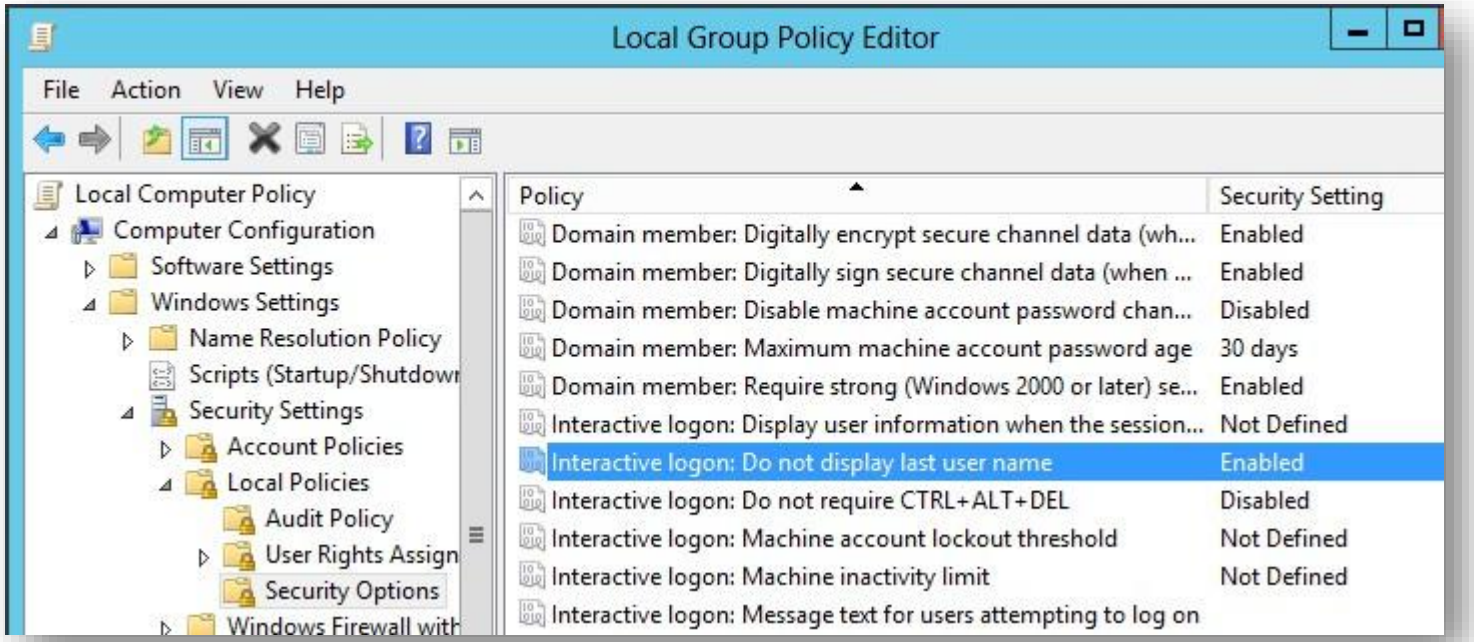
Key User Rights Assignment Settings

- Log on as a service
  - This security setting allows a security principal to log on as a service.
- Shut down the system
  - This security setting determines which users who are logged on locally to the computer can shut down the operating system using the Shut Down command.
- Take ownership of files or other objects
  - This security setting determines which users can take ownership of any securable object in the system, including Active Directory objects, files and folders, printers, registry keys, processes, and threads.

Security Policies

Computer Configuration\Policies\Windows Settings\Security Settings\Local Polices\Security Options

- Various system security settings, including settings related to User Account Control, local computer security settings.



Key Security Options Settings

- Accounts: Administrator account status
  - Whether the built-in administrator account is enabled or disabled on systems affected by the policy
- Accounts: Limit local account use of blank passwords to console logon only
  - Determines whether local accounts that are not password protected can be used to log on from locations other than the physical computer console.
  - Enabled by default
- Accounts: Rename administrator account
  - Specifies whether the built-in administrator account on systems affected by the policy should be renamed and to what

Key Security Options Settings

- Interactive logon: Do not display last signed-in
  - Prevents the logon screen from displaying the name of the last user to logon to the system
- Interactive logon: Do not require CTRL+ALT+DEL
  - Configures systems effected by the policy not to require CTRL-ALT-DEL to logon
- Interactive logon: Message text for users attempting to logon
  - Allows the administrator to specify a message displayed to users when they attempt to logon to the system
- Interactive logon: Number of previous logons to cache
  - Allows the local system to cache domain logon information so that if the DCs for the domain are not available, the user can still logon as long as their cached password matches the password they are trying to use
  - Setting the value to 0 disables logon when DCs are not available

Key Security Options Settings

- Microsoft network server: Disconnect clients when logon hours expire
  - Enabled by default
  - When a user's logon hours expire, they're account will be disconnected from file shares
- User Account Control: Use Admin Approval Mode for the built-in Administrator account
  - This policy setting controls the behavior of Admin Approval Mode for the built-in Administrator account.
  - Enabled: The built-in Administrator account uses Admin Approval Mode. By default, any operation that requires elevation of privilege will prompt the user to approve the operation.

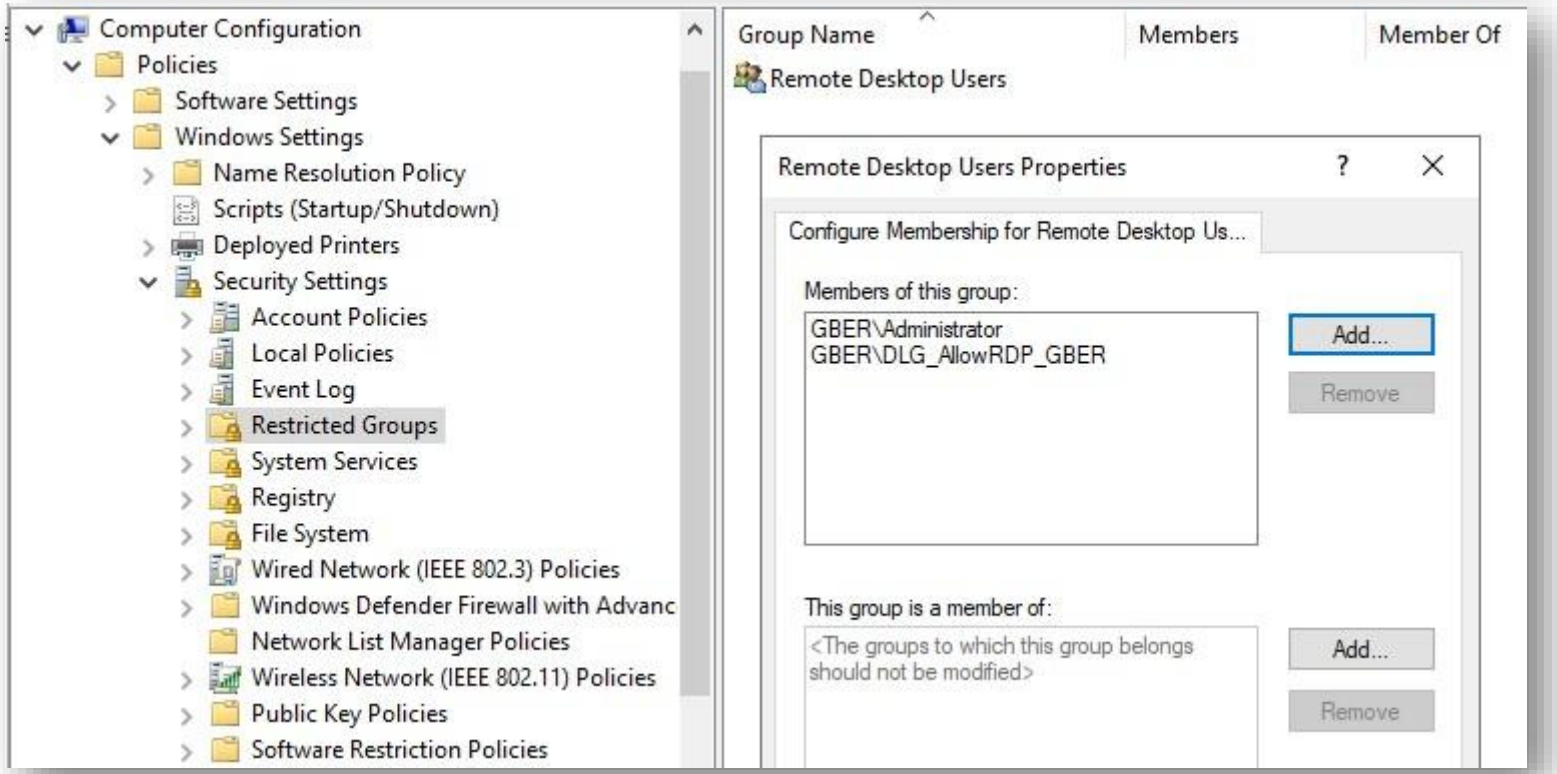
Restricted Groups

- Restricted groups allow an administrator to define group membership (members/member of) for security-sensitive (restricted) local groups.
- When a restricted group policy is enforced, any current member of a restricted group that isn't on the Members list is removed.
- Prevents users with local admin rights from changing local group membership. The members of the restricted group can only be set by the domain.
- Restricted Groups is a client configuration means and can't be used for domain groups. Restricted Groups is designed specifically to work with local groups.

Security Policies

Computer Configuration\Policies\Windows Settings\Security Settings\Restricted Groups

- Allows an administrator to specify group membership lists
  (control membership in important groups, such as Local Administrators)



Enabling Remote Desktop

(using a Restricted Groups GPO)

Steps to enable RDP using GPO.

1. Create a new GPO or modify an existing GPO.
2. Enable RDP on target systems using the following setting: Computer Config\Policies\Administrative Templates\Windows Components\Remote Desktop Services\Remote Desktop Session Host\Connections\Allow users to connect remotely using Remote Desktop Services.
   a. Filter the GPO setting on.
3. Computer Config\Policies\Windows Settings\Security Settings\Restricted Groups\
   a. Add the RDU group
   b. Add the corresponding Access Group to the RDU group in Restricted Groups.
4. Apply the GPO to the OU(s) that contain the computer objects for the target systems.

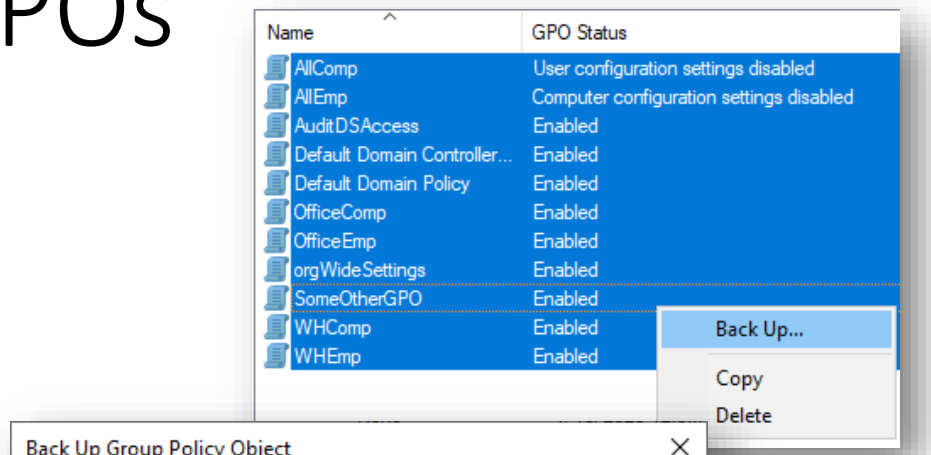
Do this…

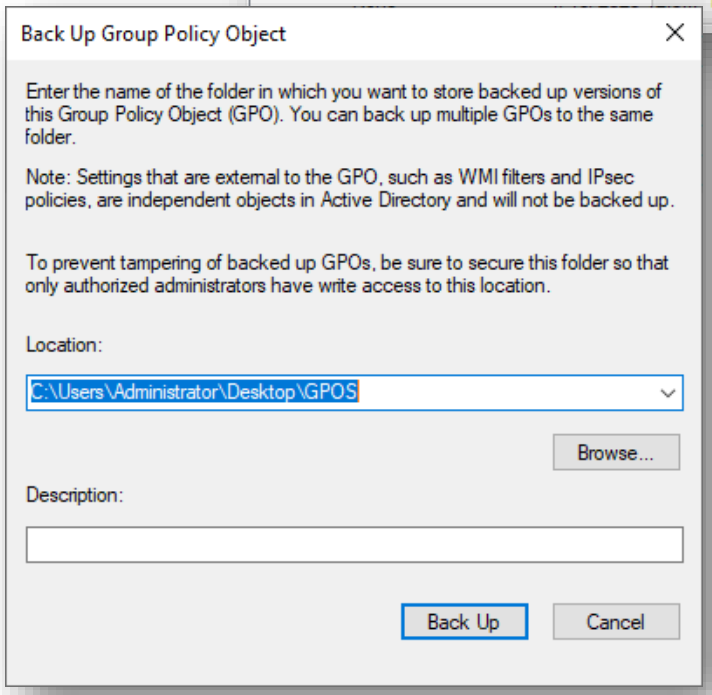
1. Create an OU called managedComputers.
    1. Put Win11-XXX and Server2-XXX computer objects in the OU
2. Create GG-<domain>-RDPUsers
3. Create DLG-<domain>-AllowRDP; nest GG-<domain>-RDPUsers in DLG-<domain>-AllowRDP
4. Create a GPO called AllowRDP.
    1. Turn on Allow users to connect remotely using Remote Desktop Services (Computer Config\Policies\Administrative Templates\Windows Components\Remote Desktop Services\Remote Desktop Session Host\Connections\Allow users to connect remotely using Remote Desktop Services).
    2. Add the local RDU to Restricted groups.
    3. Specify Domain\Administrator as a member.
    4. Specify DLG-<domain>-AllowRDP as a member.
5. Link the GPO to the managedComputers OU.
6. GPUPDATE /FORCE on both Win11/Server2
7. On the GPO, go to the Settings tab. You may want to screenshot the configured settings for the GPO.

Troubleshooting GPO

Backing up and Restoring GPOs

- Select the GPOs to back up.
- Right Click → Backup.
  - Choose a location for the backup.
  - Give an optional Description
  - The GPO is saved to a folder
- Right-Click Group Policy Object container → Manage backups.
  - Select the GPO to restore
  - Restore it.
  - You will have to recreate the links afterwards.





Troubleshooting GPO

Two tips for determining why a GPO did not apply or did not do the expected thing:

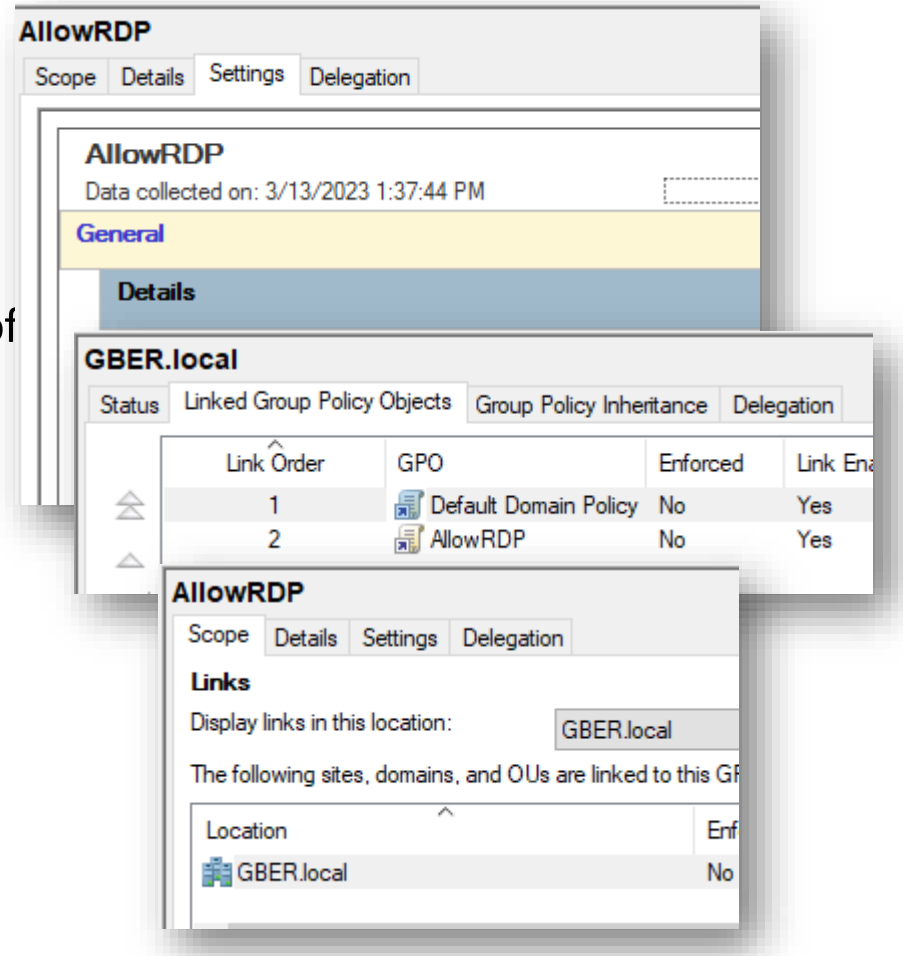
- Server-side troubleshooting
  - Look at Settings and linking in GPMC
- Client-side troubleshooting
  - Is the policy setting being set locally or by domain (iconography in GPEDIT)
  - RSoP.msc

DID YOU GPUPDATE?!

Server-side troubleshooting in GPMC

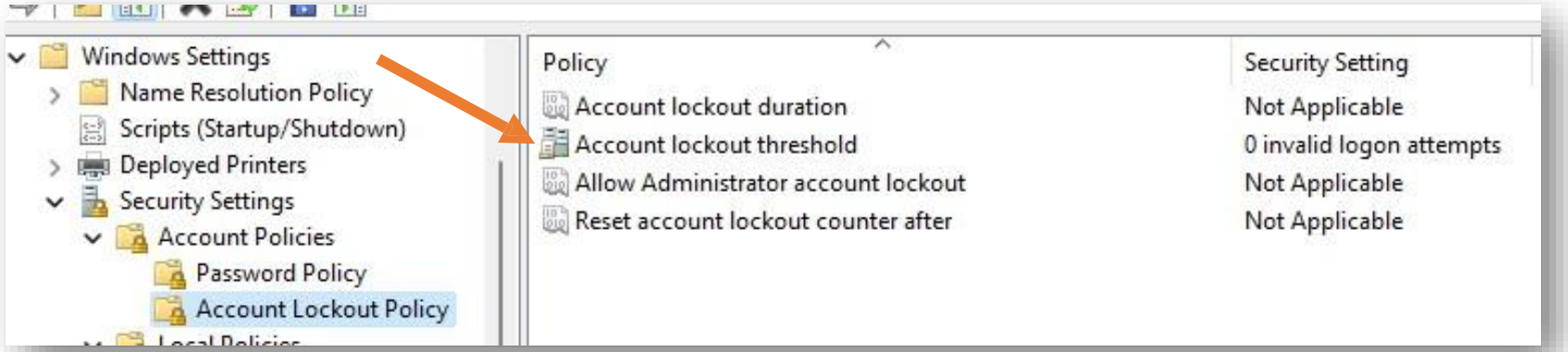
In GPMC on one of the Domains DCs

- Does the policy exist?
- Are the settings correct?
  - Settings Tab of the GPO
- Is it linked to the correct OU?
  - And is the computer/user object *in* that OU or a child of that OU.
  - Scope Tab of the GPO
  - Link and Inheritance Tabs of the OU
- In the case of possible conflicting GPOs consider…
  - L→S → D → OU → Link Order
  - Enforce
  - Block Inheritance
  - Permissions (Security Filtering/Delegation)



Client-side troubleshooting

- In GPEDIT.msc, the configured polices for the local machine can be found. A simple way (not conclusive) to know hat a GPO setting is coming from a nonlocal GPO is to look at the icon next to the setting.



- A GPO that is being applied from the domain will have a “2 computer tower” icon next to it.

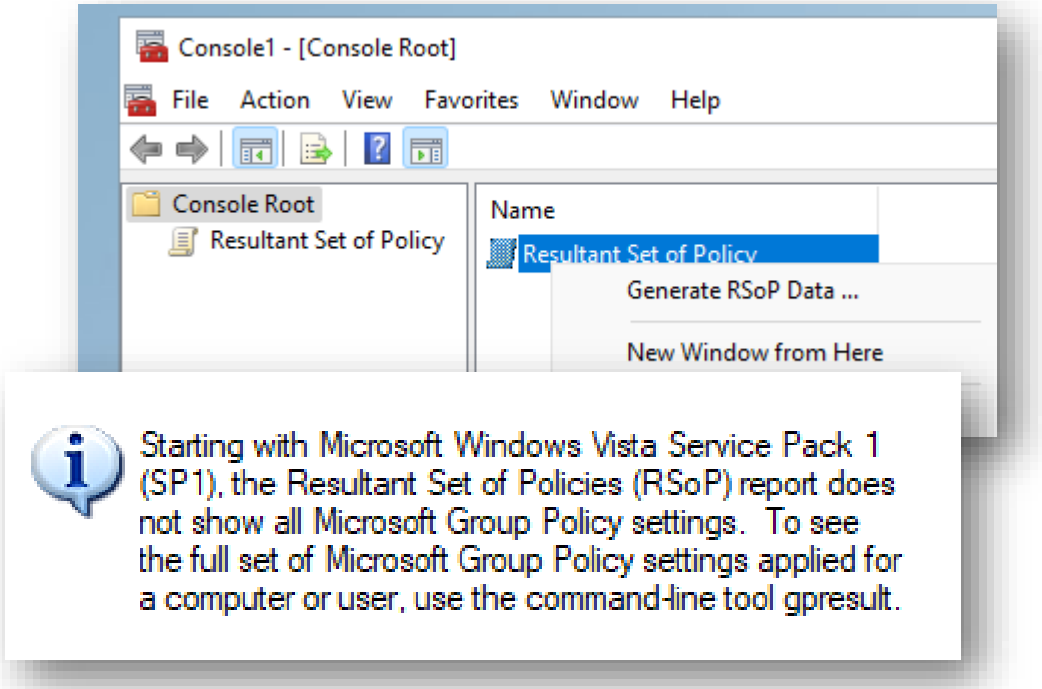
Client-side troubleshooting

(RSoP)

- A more conclusive way to troubleshoot is to use GPResult.exe or the Resultant Set of Policy MMC Snap-in. This snap-in reads the policy settings applied to the computer and creates a report.
- RSoP MMC Snap-in can:
  - Generate reports based on Machine/User/Both type polices.
  - Indicate the Name of the policy that is being applied.
  - Show the settings in each policy that is being applied.
  - Has been Deprecated.
- We'll talk about GPResult in a minute…

RSoP.msc

- You can run RSoP.msc in CMD or RUN, but doing so won't let you specify options.
- Adding RSoP to an MMC console is preferred.
- In the console select Generate RSoP Data… to enter the RSoP Wizard.
- Note that using RSoP is no longer recommended.



GPResult

- This is a command line tool that will produce the same data as RSoP and can be exported to an HTML file.

```
GPRESULT [/S system [/U username [/P [password]]]] [/SCOPE scope]
         [/USER targetusername] [/R | /V | /Z] [(/X | /H) <filename> [/F]]
```

- `/S` Specifies the Computer to query (connect to)
- `/U` and `/P` are used to run GPRESULT with different credentials than the current ones
- `/Scope` Specifies whether to display user or computer configurations (user is default)
- `/User` specifies the User to query
- `/H <filename>` outputs the results as HTML (in the current directory unless redirected)
- `/X <filename>` Saves the output of gpresult as an XML file

GPResult

```
GPRESULT [/S system [/U username [/P [password]]]] [/SCOPE scope]
         [/USER targetusername] [/R | /V | /Z] [(/X | /H) <filename> [/F]]
```

- /R displays only summary data
- /F Combined with /X or /H, causes gpresult to overwrite the contents of a file that already exists
- /V Used in place of /R, causes gpresult to use the verbose display mode (more detailed information)
- /Z Used in place of /R or /V, causes gpresult to use the Superverbose display mode

GPResult Example

- Gpresult for the computer as a non-admin user.

```
gpresult /user geoffb\administrator /scope computer /h report.html /F
```

- Remember:
  - By default, GPResult will run for the user *only* for non-admin users. Non-Admin users cannot produce RSoP for the computer (only their user account).
  - Make sure if you want to look at a specific scope of policy (user/computer) to specify it if it doesn’t appear in your results.

Get-GPO, Get-GPOReport

- Two Powershell cmdlets for viewing GPOs on a DC
- `Get-GPO –all`
  - Use this command to determine which GPOs exists on the Domain Controller
  - Then use the Name or GUID to see that specific GPO and its properties.
  - You can also use the –Domain parameter to specify your query to a specific domain (`-Domain “geoffb.local”`).
- `Get-GPO –Guid <GUID>`
  - `Get-GPO -Guid 31a09564-cd4a-4520-98fa-446a2af23b4b`
- `Get-GPO –Name <Name>`
  - `Get-GPO –Name “Remote Access GPO”`

Get-GPO, Get-GPOReport

- `Get-GPOReport`
  - Supports the `-Name`, `-GUID` and `-All` parameters like `Get-GPO`.
  - `-ReportType` can be HTML or XML
  - `-Path` is used to output the report to a path. You could also pipe to `Out-File` instead.

```
Get-GPOReport -Name “AuditDSAccess” -ReportType HTML -Path “.\Report.html”

Get-GPOReport -Name “Remote Access GPO” -ReportType HTML | Out-File “.\Report.html”
```